



Review

Advances in weld seam tracking techniques for robotic welding: A review

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ABSTRACT

Use of sensors in robotic welding for controlling the weld quality leads to replacement of manual welding operation in dangerous work environment in presence of high temperature and fumes even in small or medium scale enterprises. The seam tracking operation is very essential for extracting weld seam position which can be fed to robot controller for instructing robot along the weld seam path. The seam tracking operation can be executed by different types of sensors having their own merits and demerits. In this paper, different sensors and techniques used for seam tracking task in robotic welding have been discussed in detail. Each sensor has different method or technology of weld seam feature extraction which have been described by different authors in different ways. The chief tasks for seam tracking have been found to be weld starting and end point detection, weld edge detection, joint width measurement and weld path position determination with respect to robot co-ordinate frame. Thus sensors have a very important role in robotic welding for fully automating the system with in process real time monitoring of weld process parameters with the sensor feedback. In further discussion the practical use of different sensors in industries with a comparison of their advantages and disadvantages have been discussed. This Paper presents the role of sensors in robotic welding and a detail study of methodologies of weld seam position and geometry feature extraction by different sensors typically used for weld seam tracking.

1. Introduction

The machines those can be utilized in place of human-beings for performance of a task in terms of both decision making and physical activity are termed as robots. The utilization of robot in industries leads to the term automation which implies the introduction of robot in the manufacturing processes for both physical production processes and processing information. The wide range of application of industrial robots include material handling to an extensive variety of production applications, e.g. quality control, welding and machining. A considerable measure of research work is going on in the fields, where precision of the robot is a major factor like in robotic forming, arc welding and metal cutting etc. [1–6]. With advancement of recent manufacturing technologies, it turns into an unavoidable pattern to acknowledge automatic, adaptable and intelligentized welding producing. As is well known, welding innovation has been created and advanced from the first hand-worked specialty to the advanced systemic specialized science. Examining and simulating intelligent activity and capacity of welder's operation is critical for improvement of intelligentized mechanical welding. Therefore, simulation and understanding a welder's activities by some clever machines, e.g. industrial welding robot is one of the most vigorously researched issues in the advanced welding

technology. The use of industrial robots is essential in hazardous and poor working environment where reaching of human being is not possible like in ship building application, nuclear power plants, maintenance and repair works etc. Mechanical robots and motorized hardware have gotten to be crucial for modern welding for high-volume profitability since manual welding yields low production rates. For small or medium production volumes, automated generation yields the best cost per unit execution when contrasted with manual mechanization. Robotic welding frameworks bring different favorable circumstances, for example, make strides in efficiency, weld quality, adaptability and workspace use, and diminishes work costs in addition to focused unit cost [7–10]. Be that as it may, welding robots basically work in “teach and playback” mode in which they have no enough adaptability, particularly in welding process. Since welding as a process experiences numerous factors such as the mistakes of pre-machining, fitting of work-piece, and in-process defects, which will bring about variation in weld gap, the welding techniques must be balanced as per these progressions. Robots in instruct and playback mode have no such capacities and typically weld a weldment with bunches of defects and poor penetration. In this way, it is important to build up an automatic control framework for those welding robots working in instruct and playback mode [11,12]. A genuinely mechanical welding framework

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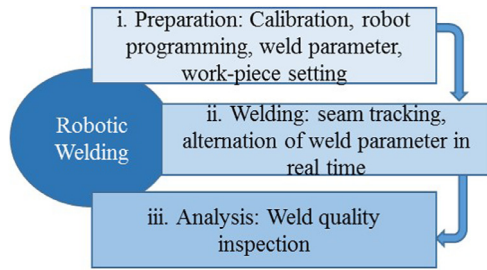


Fig. 1. Steps of robotic welding.

ought to have three foremost parts: sensors, which identify the state of welding procedure; a model of process, which gives the relationship between the welding factors and the geometry of weld bead; a control framework, which assesses the sensor information and changes the welding procedure factors utilizing the relationship in the process model [13–15].

1.1. Robotic welding

Welding process has been one of the most integral part of modern manufacturing process. Now-a days, the utilization of industrial robots is one of the chief sign of contemporary welding [16]. The integration of robot with welding process has advantages like: (1) The robotic welding can be very useful in terms of both cost and quality for small and medium production volume. (2) Consistency in performance by maintaining the quality for a long time. (3) Makes the process quite flexible and also can be utilized for other modern manufacturing processes. (4) Adaptation of variation in production line with variation in production volume (Easy adaptation to change in design and working environment). In automated welding generally three stages are there: (i) preparatory stage, (ii) welding and (iii) analysis as shown in Fig. 1 [17].

1.2. Need of sensor in robotic welding

To move the welding torch along the weld seam path, teach pendant used to be utilized in traditional robotic welding system. In addition to this, programming for setting of weld parameter, starting and extinguishing of arc and controlling of shielding gas and use of robot controller for smooth welding have to be performed simultaneously. But this often results in to error in torch positioning and unsatisfactory weld quality. These issues can be handled easily by the use of sensors for seam tracking and weld pool inspection [18]. There is an increase in demand of concept of intelligent robotic welding due to variation of part design, just in time and CAD customized flexible production approach. Sensor integration with the robot manipulator can improve the welding process in an intelligent manner [19]. The different aspects of automated robotic welding are shown in Fig. 2.

The introduction of sensors in production processes leads to a progressive step towards achievement of expanded adaptability with respect to volume of production, product variation and life cycle of product. The sensors in production processes can be utilized for optimization and process control of diverse machines and different

machine tools. The sensors can also be used to expand the adaptability of the robot framework and empower the utilization of robots in irregular production which leads to recognition of various issues that must be understood effectively to coordinate sensors and enhance effectiveness of the system. But it has been observed that the actual use of sensors in welding processes is quite low except for high quality and tolerance requirements [20]. The primary use of sensors in welding operation is to control the weld parameter and the actions of robot. Pires et al. [17] described about development of seam tracking system as described in Fig. 3.

2. Sensors utilized for seam tracking

The sensors utilized for trajectory control of robot path in welding are generally based on arc sensing and optical sensing. But other sensors are also available like infrared sensing, ultrasound sensors, electromagnetic sensors etc. having their own method of characteristic feature extraction but have not been utilized in industries in wide range. The sensors can be utilized in different types of robotic welding.

Again various type of sensors are used in robotic welding for performing seam tracking and are represented in Fig. 4 according to their working principle.

2.1. Arc sensor for weld seam tracking

The arc sensor is typically designed for metal arc welding process and depends on a detecting technique which uses a key arc property, i.e. arc voltage and current features change according to the height of torch comprising of arc length and wire extension changes. The very reality of the arc fundamentally being used with advantages of its status as a modest and basic method, great continuous operability, and little defencelessness to the impact of wire electrode twisting, have made this approach famously material to different sorts of programmed welders and robots [21]. A general process of seam tracking by arc sensor is described in Fig. 5.

2.1.1. Developments models for arc sensor

Past studies developed mathematical expression showing relation between the torch to work-piece distance and the welding parameters like voltage current, wire feed rate etc. Julian [22] developed mathematical model for arc sensor considering its static and dynamic properties. In this model, the static properties, equilibrium conditions of wire feeding system and electric circuit have been utilized.

First an equation has been established utilizing the electric circuit parameters as shown in Fig. 6. Where, U represents the arc voltage, V_f is the wire feed rate, I is the weld current and H is the torch to work-piece distance. Based on the wire feeding system condition then another equation for melting rate has been established in terms of melting rate devoted by arc and the wire extension. Both the equations are combined to develop expression for torch work-piece distance. Experiments have been performed to compute the coefficients by regression analysis and the generalized equation developed is shown in Eq. (1).

$$H = (a_1 + b_1 V_f)I + (a_2 + b_2 V_f) \quad (1)$$

As because the arc sensor either rotates or oscillates at the time of

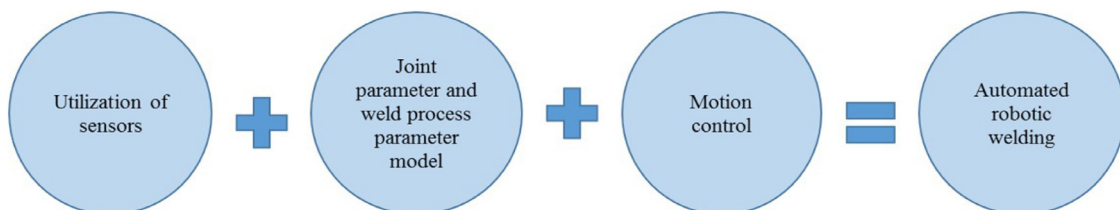


Fig. 2. Aspects of robotic welding.

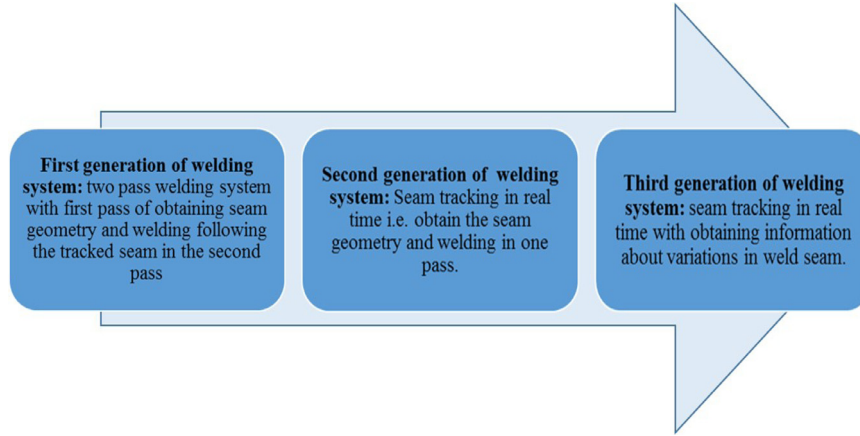


Fig. 3. Seam tracking system development.

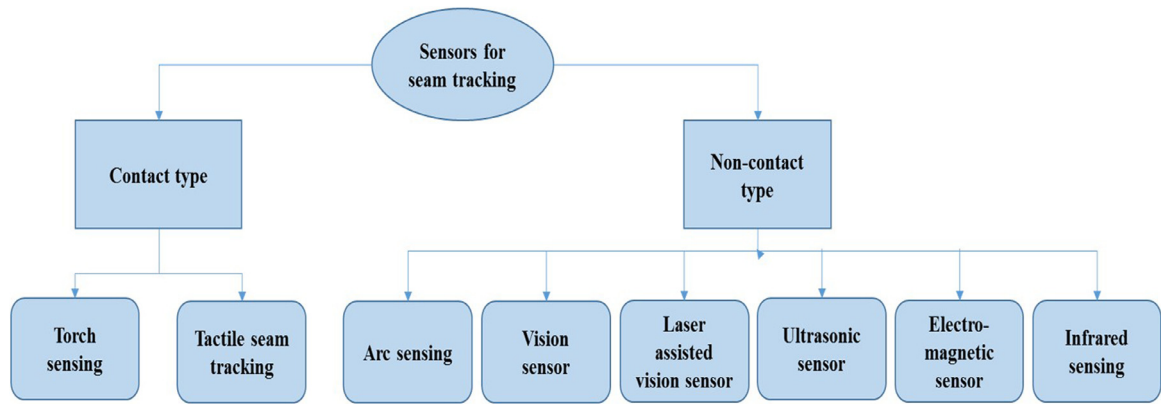


Fig. 4. Types of sensors.

welding its dynamic properties have been obtained by converting the equations of static model in to dynamic equations. By Laplace transform the equations for dynamic feature of power source or relationship among welding current and output voltage have been developed and finally from the electrical circuit block diagram the transfer function of arc sensor have been developed. With the help of Bode diagram the results are then expressed for validating the mathematical model. These models are typically useful for development, design and experimentation of arc sensors. Ushio and Mao [21] proposed a dynamic wire melting electrode model which is further used for developing model for an arc sensor typically utilized for DC metal inert gas welding in open arc mode to understand arc sensor more quantitatively. The melting of electrode wire is fundamentally restrained by Joule heat at the wire extension tip and heating by arc heat conduction. Based on these, the

mathematical model for dynamic melting features of wire have been evaluated as Eq. (2).

$$\frac{dL_e}{dt} = V_f - \frac{AI}{1 - BJ_e} \quad (2)$$

where $A = \phi/S_0\gamma_0(H_m + H_{latent} + C_{pl}\Delta T + a_0)$

$B = a_1/(H_m + H_{latent} + C_{pl}\Delta T + a_0)$, where A , B are the wire melting model coefficients, L_e is the wire extension length, V_f is the wire feed rate, I is the welding current, J_e is the Joule heat quantity at wire tip, ϕ is the equivalent wire melting voltage of arc, S_0 is the cross sectional area of wire at room temperature, S_0 is the wire density at room temperature, H_m is the electrode wire heat quantity at melting point, C_{pl} is the specific heat of electrode wire in liquid state, ΔT is the difference of temperature between temperature at contact tip and at melting point

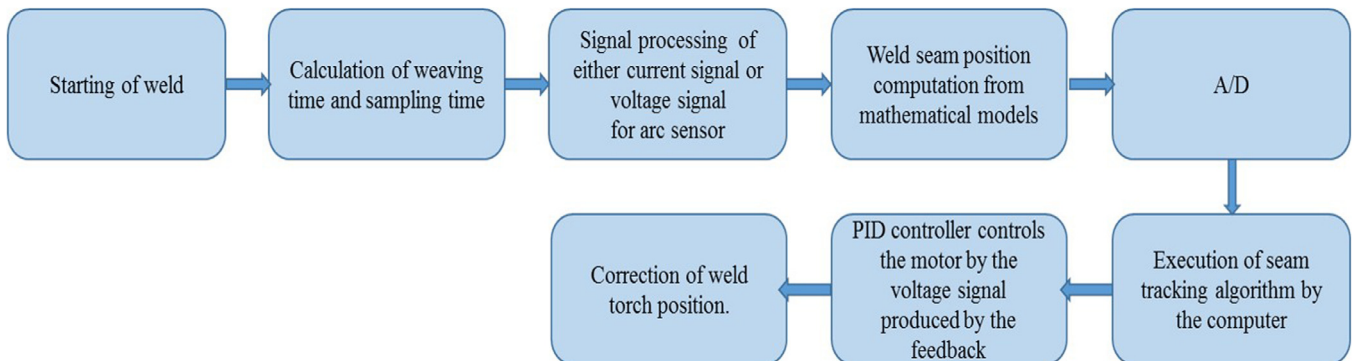


Fig. 5. Method of seam tracking followed by arc sensor.

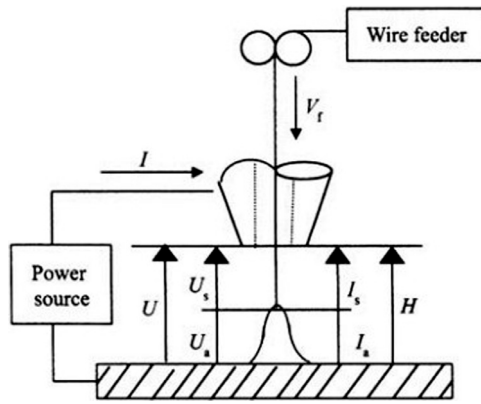


Fig. 6. Model of arc sensing system based on static properties [22].

and a_0 , a_1 are the coefficients used in fitting of Joule heat quantity of electrode wire.

From Eq. (2), the static wire melting model has also been derived. Then to govern the expression for relation between torch height, welding voltage and current, a non-linear mathematical model has been developed for DC MIG welding in open arc mode. The linearization of the above arc sensor model has also been performed with assumption of little variation of variables around a point of consideration in operation.

2.1.2. Seam tracking algorithm and control module for arc sensor

There are several past works to develop a control module for the weld seam path planning and control of weld parameters for getting quality weld. Moon et al. [23] studied mainly on the development of arc sensing algorithm considering the effect of change of tip to work-piece distance at the groove edge resulting in to a stable tracking in applications of pipe line welding. The constant current mode has been considered for lead wire to track the weld seam, modified constant current has been considered for trail wire to improve the stability of arc and the weld quality. For this, a modified moving average algorithm has been proposed for the signal processing based on parameters weaving time and voltage signal. Friedenfolk and Bolmsjö [24] proposed a 6D seam tracking algorithm based on position, pitch and trajectory tangent vector control. The position control has been aimed to calculate the position errors by utilizing differential method for computation of position. Then by curve fitting the trajectory has been estimated. Finally for pitch control, distance between tool centre point to seam wall and direction of current have been determined for on line control of orientation and position of torch. Jeong et al. [25] presented a high speed rotating arc sensor based on microprocessor based seam tracking controller system. A software filter algorithm using moving average method and an exponential smoothing method applied to welding current have been used in microprocessor for reducing the effect of noise. Depending on the value of average current at every interval of four phases per one rotating cycle, the seam tracking algorithm has been formulated. The weld seam direction has been determined after estimation of the difference of average current at right and left points. Chen et al. [26] proposed a Human Simulated Intelligent Control (HSIC) controller based on HSIC prototype algorithm having two level of control structure i.e. parameter level and operation level. When compared with other controller like PID controller, neural network controller or Fuzzy controller, the proposed controller was found to be more simple and quick. But from the simulation results, it has been found that the Fuzzy controller's performance is better in some aspects. Yanfeng et al. [27] proposed hybrid control method based on back stepping and neural network algorithm for mobile robot slider and mobile platform typically used in ship building applications. First, the back stepping controller has been designed for the posture tracking error taking control parameters of robot like reference welding velocity and angular position for the mobile platform and slider. Then stability

Table 1
Papers with utilized methodology and different parameters for arc sensor.

Year	Authors	A	B	C	D	E				F	
						Wire feed rate	Welding speed	Current	Voltage		CTWD
2001	P. Julian [22]	Mathematical modelling based on static and dynamic properties	–	Pulsed MIG/lap joint and I-butt joint	✓	–	✓	✓	✓	–	–
2001	Jeong et al.del. [25].	Seam tracking algorithm based on average current value	4.5	DC MIG/V-groove	–	✓	✓	✓	✓	–	Thin plate welding
2004	Fridenfolk et al.del. [24].	6D seam tracking system	–	V-groove	✓	–	✓	–	–	–	Manufacturing of large ships, cruisers
2009	Moon et al.del. [23].	Modified moving average algorithm	–	GMAW	–	–	✓	✓	–	–	Pipe line welding
2009	Ushio and Mao [21]	First a non- linear and then a linear model for arc sensor	–	DC MIG/MAG	✓	✓	✓	–	–	✓	–
2011	Chen et al.del. [26].	Human Simulated Intelligent Control controller with tracking error as the input.	–	–	–	–	–	–	–	–	–
2011	Yanfeng et al.del. [27].	Back-stepping method for designing kinematical model and neural network to recognize the kinematic model	–	–	–	–	–	–	–	–	Ship building

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/Weld joint, D- Weld parameters, E- Sensor parameters, F- Applications



Fig. 7. FD-TR TIG arc sensor by Daihen [28].

analysis has been done utilizing the Liapunov function. Using the velocity commands acquired from non-linear feedback control law, the PID controller has been controlled the motor. The robot kinematic model finally determines the robust posture with the help of robot kinematic model through data obtained from the encoder attached to the motor. A back propagation neural network has been designed to adjust the control gains in real time. The summary of papers studied for arc sensors are listed in Table 1.

2.1.3. Arc sensors available in market

There are products of arc sensor from Daihen company called FD-TR TIG Arc Sensor as shown in Fig. 7. The welding torch of MIG setup is traveled by the FD-TR arc sensor which measures the arc voltage during welding. By this, feedback is received through the tungsten electrode in the torch and constant arc length is maintained. The real time seam tracking operation of the FD-TR system is very easy and simple through the dedicated menu in the teach pendant [28]. Similarly arc sensors are also available from other companies like ComARC from Yaskawa Motoman, Arc seam sensing from company igm and POWER-TRACTM from SERVO-ROBOT etc.

2.2. Vision sensor for seam tracking

At present, modern welding robots are customized through 'teach and playback' techniques by human administrators. It can take a huge measure of time and cost to program ways and refine welding parameters for every part. To bring the advantages of automated welding to these ventures, it must turn out to be more adaptable. Computer vision can be utilized to recognize and find the position of welding creases which can be used to arrange a way to weld the parts naturally. This would ease one of the real downsides of mechanical welding [29]. A general process of seam tracking with vision sensor is shown in Fig. 8.

2.2.1. Vision sensor seam tracking system specification

Ge et al. [29] made use of 256 MB memory and 800 MHz processor PIII CPU and Charged Coupled Device (CCD) camera with proper narrow band pass filter for removing noise of arc in the seam tracking system. Ma et al. [30] utilized a CCD camera, a wide-band filter of transmission 590 nm to 680 nm and peak concentration rate of 4%, neural density mirror of attenuation 90% and reflective mirror for construction of visual sensor for metal inert gas welding of LF6 Aluminium alloy of 2 mm thickness. Xu et al. [31] described the configuration of visual sensor as a composition of three roller screws, a camera mounted about 150 mm above welding torch tip to capture images, a programmable logic controller (PLC) working as the control computer for adjusting the torch orientation and a touch panel

computer to compute the error between torch and seam for CO₂ gas shielded arc welding of 0.5 mm width butt weld. Nele et al. [32] described major elements of image acquisition system as color CCD camera with a resolution of 640 x 480 pixels assisted with 90 LEDs arranged on a PCB for proper illumination to acquire images in real time. For seam tracking in real time, LabVIEW software platform has been chosen. Xu et al. [33] performed spectrum analysis of the GMAW of 2.5 mm Q235 steel to find out the appropriate optical filter for removing disturbances due to the presence of arc light. From the experiment it has been found that optical filter having central wave length 660 nm would be most suitable for welding under arc light. Whereas, Chen et al. [34] found narrow band filter of central wave length 630 nm and neutral glass slices of 10% + 30% + 50% would be suitable for capturing images in welding of Aluminium alloys having light spectrum distribution 580 nm to 720 nm. Gao et al. [35] utilized a narrow band composite filter consisting of a narrow band filter of central wavelength 610 nm and band width 20 nm and a neutral filter for a specimen of mild steel Q235. Zhou et al. [36] introduced a two CCD camera based seam tracking system which is mounted on opposite sides of outer hollow shaft for capturing the both seam image and weld pool image. With the help of electromagnetism air valve and two air cylinders, the sensors can load or unload light filters. Shen et al. [37] designed a double layer filter system consisting of wide band filter and two dimmer glass to facilitate various light intensity between seams and weld pool for Aluminium alloy welding as shown in Fig. 9.

2.2.2. Calibration of vision sensor

Vision sensor or camera calibration is performed to show the relationship between vision sensor frame features and real world co-ordinates with respect to robot frame co-ordinates. To remove the systematic errors and to enhance the precision of measurement camera calibration is an important step. By camera calibration, its intrinsic and extrinsic parameters can be obtained very accurately. Xu et al. [33] applied MATLAB camera calibration tool boxes for this purpose. As shown in Fig. 10, the black and white planar check board pattern filled with 5 × 5 mm planes have been used for camera calibration.

Again calibration is also required in terms of conversion of image plane co-ordinates relative to absolute co-ordinate system. The relationship between the image and absolute co-ordinate system has been expressed as Eqs. (3) and (4).

$$x_{real} = \{-b + [b^2 - 4k(x_{image} - x_{real})]^{1/2}\} \times d_{real}/2k \quad (3)$$

where, (x_{image}, y_{image}) is the image plane co-ordinates, (x_{real}, y_{real}) is the absolute co-ordinates values, (0,0) is the origin of absolute co-ordinate system on the image plane co-ordinate system, k and k', b and b' are the scale factor and intercept, d_{real} is the interval of planes [33,37–39] proposed a triangulation methodology for transformation of 2D pixel co-ordinate values of a point in two or more images to 3D real world values typically for robot path planning with two camera in vision sensor. This methodology follows three steps: (1) camera calibration (2) radial lens distortion removal (3) computation of final 3D Cartesian co-ordinate position of weld seam.

The first step of this involves the use of a calibration grid (30 × 30 mm) containing red color filter for defining origin and blur color filter for defining the calibration area. The attributes of camera have been computed utilizing the proposed automated grid detection algorithm along with MATLAB calibration toolbox. Utilizing the right hand rule, conventions of all the co-ordinate frames have been established. Finally the hand eye transformation is obtained from the robot-world transformation [40,41]. In the second step the distortion free pixel co-ordinates are obtained with normalized image co-ordinates to remove the effect of radial distortion of lens. Then in the third step the optimized triangulation method has been adopted to obtain the 3D Cartesian co-ordinates of point in images [39].

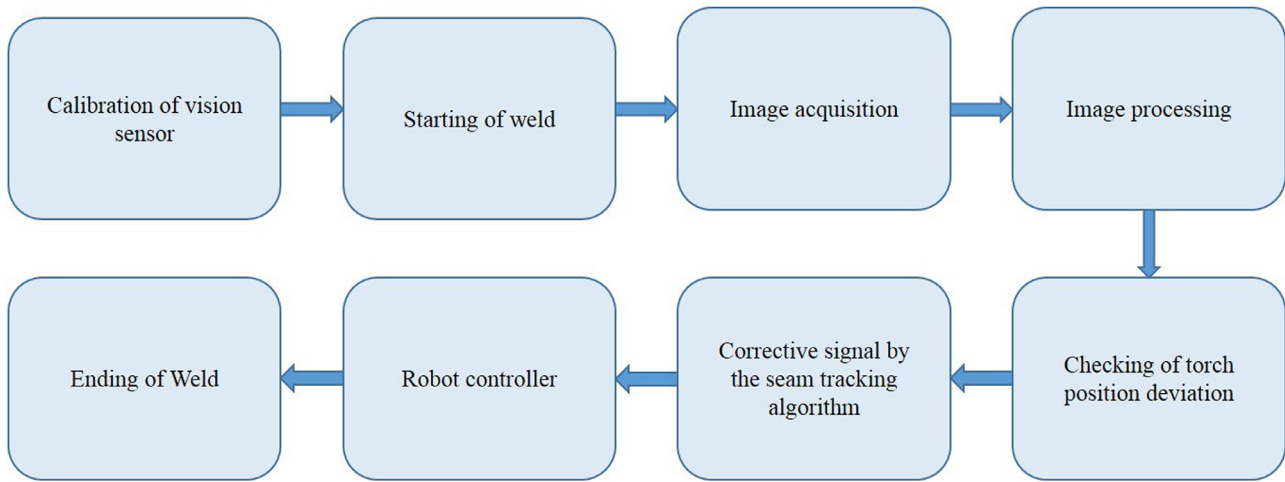


Fig. 8. Seam tracking process by vision sensor.

2.2.3. Image acquisition and start of weld

After calibration the next step is image acquisition i.e. capturing of weld seam image by the vision sensor and selection of image with auto finding of weld start point. According to the pulse frequency of current, the image capturing time is fixed [30]. The software platform used for seam tracking should enable to record the image at a particular frame rate and save the captured video. LabVIEW software package has been found very suitable for image acquisition in automatic mode. Because of the effect of welding current, the arc light undergo the bright to dim cycle. Therefore, the image should be acquired when the current is minimum for GTAW. But as for GMAW the current pulse frequency is very high, it is not possible to capture image during only base current period. Therefore dimmer filter system as mentioned in design of vision sensor are generally used for GMAW at a frame rate of 25fps. Before going for image processing, it is necessary to select useful images as the acquisition process is very much affected by spattering, droplet transfer, fuming and associated noises. Depending on the maximum average

intensity gradient value of the edge points of a line drawn through every image, the images have been selected for each second [32,33,42]. At the beginning of seam tracking, it is necessary to find the starting position of torch. With the help of the image signals, the work-piece location can be obtained to orient the torch accurately at its initial position. Nele et al. [32] proposed a template matching procedure to obtain the initial position of torch for seam tracking automatically. The point of intersection of lower edge of work-piece plate with the edges of seam can be considered as the starting position of welding. Micallef et al. [43] defined the initial and final position of weld seam as the maximum or minimum pixel values of x-axis or y-axis in image plane depending upon the distance between the corresponding minimum and maximum pixel values.

2.2.4. Image processing

After image acquisition and selection of images the next step is to extract the weld seam features by image processing algorithm. Ge et al.

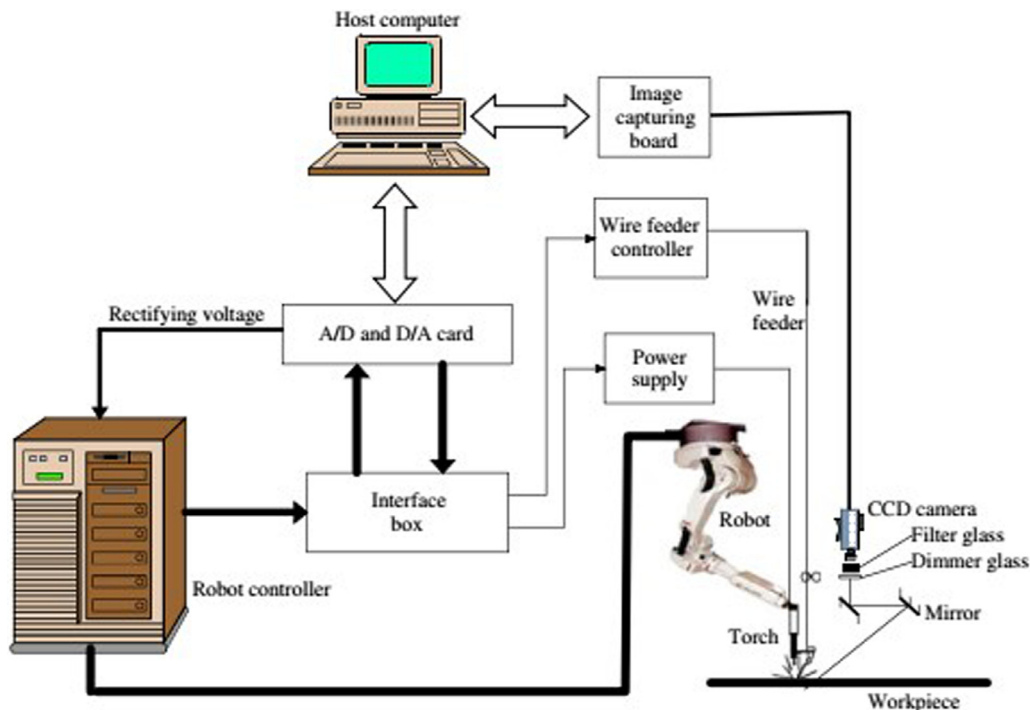


Fig. 9. Seam tracking system in real time with vision sensor [37].

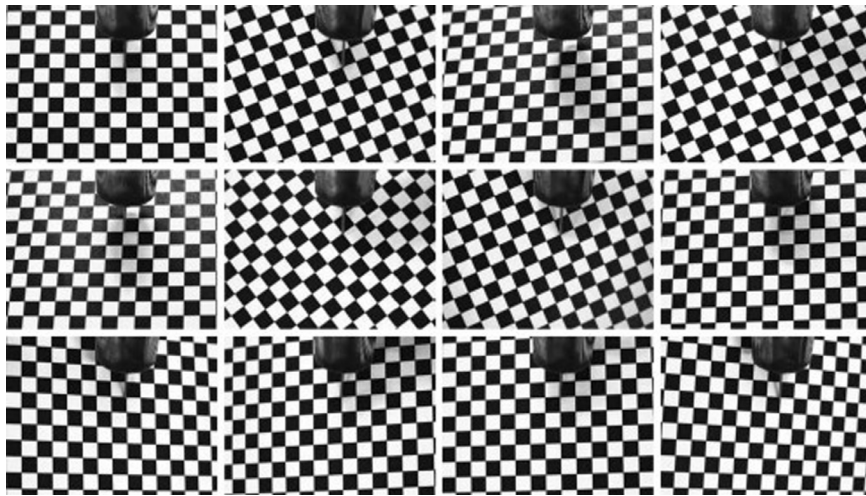


Fig. 10. Camera calibration images [33].

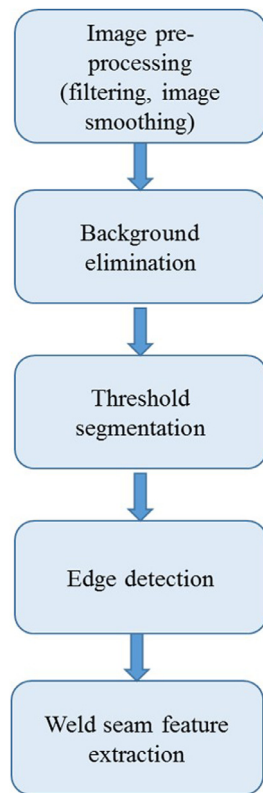


Fig. 11. Image processing algorithm followed by vision sensor.

[29] proposed the molten pool area image processing algorithm involving steps like utilization of diagonal differencing operator to compute the pixel values in image, binarization of image by considering the statistical property of pixel, edge extraction defining two areas in image i.e. molten pool area and not welded seam area and extraction of feature points of the molten pool area by a loop. Generally the image processing for robotic welding involve steps as shown in Fig. 11.

2.2.4.1. Image pre-processing. The image pre-processing involves, finding of noise points by analysis of gray level and then removal of these noise by different filtering method like median filtering for removing salt and pepper noise and impulse noise, average filtering, morphological filtering typically for different noises around weld pool and the seam [33,35–37,44,45]. Two types of filters can be utilized for

highly noisy and blurred images i.e. (i) linear or convolution filtering including low pass filter and high pass filter (for low frequency noises) resulting sharper images (ii) non-linear or non-convolution filters (for random noises) [39,46].

2.2.4.2. Back ground elimination. The next phase is elimination of background or threshold segmentation for segmenting the object in to two regions i.e. object and the background. The initial step is to create the chart for gray level distribution. For most of the part, the proper value of threshold is given by the dip, separating the two groups of gray level information, which will be evident on the histogram. The least gray level estimation of the dip (set apart as V) is the most appropriate value for threshold to adopt [46,47]. Chen et al. [34] proposed an improved adaptive threshold algorithm for solving the uneven background of the weld pool image segmentation. Dinaham and Fang [39] applied Hough transform for following the sliding search windows, which can be further used for comparing the pixel intensity across the Hough line as shown in Fig. 12.

2.2.4.3. Edge detection technique. This is the most important phase of image processing method. Under arc environment, Hough transform has been found suitable for extracting seam lines very efficiently [31]. Xu et al. [33,38] have given a comparative study of different edge detection technique for GMAW as shown in Fig. 13 and proposed an improved Canny edge detection technique resulting an image with local smoothing effect.

Gao et al. [35] found Hough transform suitable for edge detection technique in GMAW. For off-line seam tracking method, a comparative study of different edge detection techniques has been done and it has been found that Prewitt operator was more suitable with the auto-selecting threshold value method when compared with Robert, Laplacian, Canny and Sobel edge detection techniques. Roberts's operator has been found suitable for weld seam edge detection typically for Gas tungsten arc welding for aluminium alloy work-pieces [37,44]. Dinham and Fang [48] proposed an adaptive line growing algorithm for

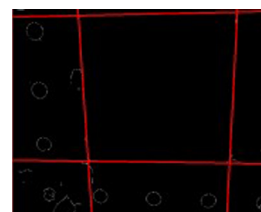


Fig. 12. Image after Hough transformation [39].

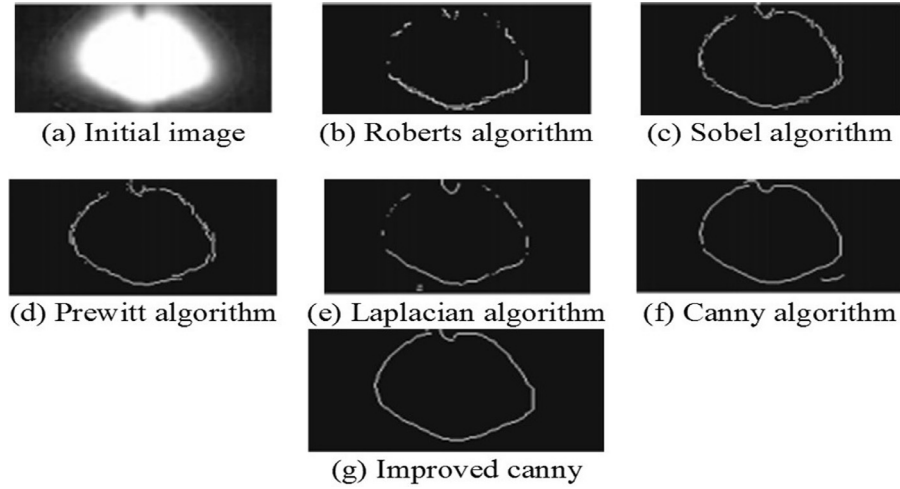


Fig. 13. Edge detection techniques comparison [38].

detecting the weld joint for fillet type of joint of any shape. This algorithm is based on the assumption that compared to work-pieces on either side of the joint, the weld seam is darker.

2.2.4.4. Weld seam feature extraction. This is the last phase of image processing by which the weld seam features for seam tracking operation have been abstracted. The purpose of every image processing technique is to obtain co-ordinates of the centre line of the weld seam and the co-ordinates of the centre of the weld pool from the weld image. The centre line of the weld seam can be obtained by the average of the two equations of the left or upper and right or lower edge and by knowing the co-ordinates of weld pool centre, the deviation can be calculated [33,37,38,44]. Suppose the expression for upper and lower edge curves are given as,

$$\begin{cases} f_1(x) = k_1x + b_1 \\ f_2(x) = k_2x + b_2 \end{cases} \quad (4)$$

Then the expression for seam centre function $f_3(x)$ can be computed by eq.(6),

$$f_3(x) = \frac{f_1(x) + f_2(x)}{2} = \frac{k_1 + k_2}{2}x + \frac{b_1 + b_2}{2} \quad (5)$$

Thinning is the process by which the seam centre line can be obtained from the binary valued image regions. This process is typically used for very narrow weld where edge detection is not possible [36]. For GTAW after edge detection, skeleton thinning and fitting of edge seam can be obtained by non-linear square method as shown in Fig. 14. The seam centre function can be computed from the average of seam up-edge function and seam down-edge function for the window of image taken for weld seam. Then the co-ordinates of torch point projection have

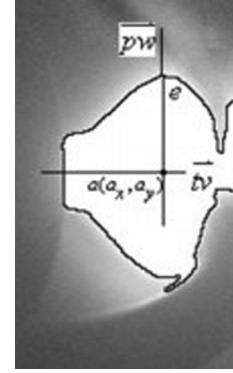


Fig. 15. Final result of image processing of the window of image taken for weld pool centre [37].

been computed from the window of the same image taken for weld pool orientation of tungsten electrode and are described with Fig. 15. From these values the offset value for torch can be achieved.

As shown in Fig. 15, $\overline{pw}$ represents the tungsten electrode orientation, $\overline{pw}$ is the line with the point “e”, where arc is widest and point a represents the centre of weld pool [37]. Shi et al. [49] developed a seam tracking algorithm for detection of weld seam features from a single image. First parallel edges have been searched in the local area of weld seam and then the leftover edges are found by iteration from the end points of each edge with the help of shift window Li et al. [50] proposed a fast Gray Projection Integral (GPI) algorithm or modified GPI algorithm which is different from other line detection technique and is a quite new and fast method for detecting straight weld seam. Gao and Na

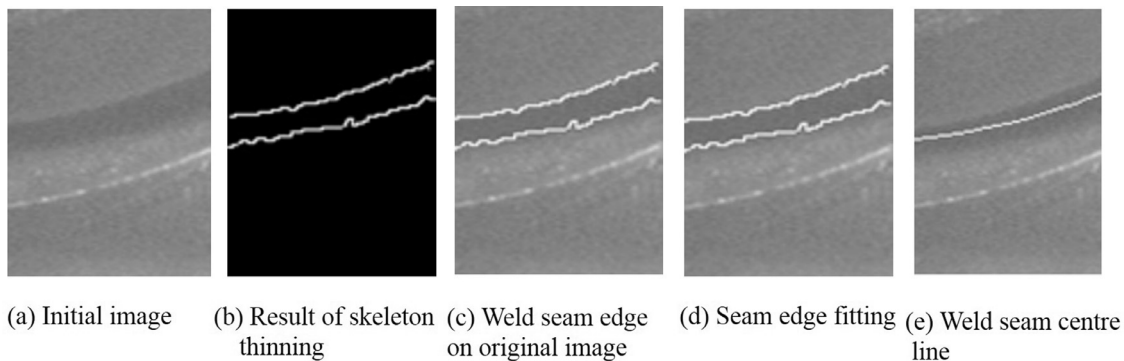


Fig. 14. Image processing of the window of the image taken for seam path [37].

[51] proposed a Kalman filter algorithm for establishing weld position equations typically for GTAW. The equations of weld position are based up on the centroid position of the weld pool images which are treated as the basic measure value or the Eigen value. Kuo and Wu [46] considered three types of weld seam path i.e. straight line, kinked straight line and curved line. Edge detection technique and fuzzy theory have been applied to obtain the final weld position co-ordinates. Fuzzy hypothesis and edge-operator location strategies are utilized to create the gray level characteristic elements and the membership function of the captured image.

2.2.5. Controller design

The weld seam position values obtained after image processing are then utilized by the control module of seam tracking system for guiding the robot along the weld seam path. Ma et al. [30] analyzed fuzzy proportional-integral-derivative (PID) controller for seam tracking through consideration of various offset values. The auto-regressive with exogenous input model (ARX model) has been designed to identify the relationship between offset and the rectifying voltage. For the PID controller the offset and the change in offset are the input parameters and PID's parameters are its output. Xu et al. [31] designed a decision controller through the PLC to control the stepper motor so that the tracking or aligning errors can be removed. According to the input to PLC, the method of visual control system has been decided. The method consist of three steps like preparing, welding and returning. According to this, three switch functions have been designed for aligning, reference setting and tracking. Zhou et al. [36] proposed a tracking controller to guide the torch along the weld seam at a constant velocity. When the position error is zero, the robot is controlled to guide the torch along the local seam tangent direction with feed forward speed. When the position error is not zero, it is compensated by displacing the torch along the vertical axis toward seam with an extra feedback speed. Shen et al. [37] proposed a simple PID controller after conducting a lot of experiments. For controller, offset value and rectifying voltage have been considered as input and output parameters respectively. Jinqiang et al. [35] also developed a PID controller with deviation of root gap centre to the torch position as input parameter and movement of the torch in a direction vertical to welding direction as output parameter.

Ting et al. [52] designed a controller based on mobile robot dynamics for seam tracking. For the cross slider and wheels of mobile robot control in co-ordination, Fuzzy-Gaussian neural network and self-turning fuzzy controller have been utilized. Chen et al. [34] constructed a nonlinear self-corrected controller with geometric features of the weld pool as the input parameter and the corrective voltage as the output parameter. To compensate the variation in weld gap a feed module has been engaged with the controller. The summary of papers studied for vision sensors are listed in Table 2.

2.2.6. Vision sensors available in market

The MotoSight 2D Global Edition sensor system available from company Motoman, can be used to obtain solutions for complex vision application and shown in Fig. 16. It is generally utilized for guiding robot, part identification and inspection, assembly and welding etc. The vision system is programmed by Cognex EasyBuilder programming software [53].

Similarly vision sensors or typically CCD cameras are available from companies like KEYENCE, Cognex, Fanuc iRvision and ISRA Vision etc. These sensors has to be used with proper housing or cover for protection against spatter and dust. Also different optical filters are used with the vision sensors for removing disturbances due to the arc light during welding

2.3. Laser vision sensor for seam tracking

In the vision sensor using only CCD camera for seam tracking becomes very complicated as the algorithms proposed for extracting the

weld seam position from the two dimensional image plane are not so accurate and time consuming. Again the measurement of weld groove depth, weld gap and detail shape condition are quite difficult. Therefore, many researchers have investigated vision sensor with structured light emitter i.e. laser diode to come with a solution for the above stated problems. Wilson [54] described the utilization of laser sensor in various industries, mainly automobile industries. The first installation was done by Ford for doing the welding operation on transit under body.

Zaeh and Vogl [55] discussed the application of laser projection interaction with industrial robots and their programming methods and mechanisms for editing typically in hand craft companies or small or medium sized enterprises. A general process of seam tracking utilizing vision sensor is shown in Fig. 17.

2.3.1. Structure and working principle of laser vision sensor

A basic laser vision sensor consists of three parts i.e. laser diode, CCD camera and filter. The laser diode may produce a stripe or dot which would be scanned by the camera. The CCD camera is always fixed at an angle to the laser to capture properly the projection of laser on work-piece [56]. Gu et al. [57] used a laser sensor based seam tracking system consisting of a stripe laser, a CCD camera and a filter. The CCD camera is equipped with a JC-629 industrial camera with focal length of 16 mm. The laser wavelength of stripe laser is of 650 nm which is selected depending up on the arc light spectrum for welding of steel. The optical filter with band width centred at 647.7 nm and transmissivity of 35% is placed in front of CCD camera to filter out all the noise and disturbance from arc light.

Agapiou et al. [58] described the selection of laser sensor based on analysis of MIG spectrum. The proposed setup for laser vision sensor consists of diode laser of output 30 mW, linear CCD camera of 1024 pixels which transfers the digital information through RS-422 interface to a personal computer, oscillating mirror equipped with an optical encoder and an oscilloscope in the range of 5–20 Hz. The spectra has been taken for welding currents 100, 120, 140 and 160 A (21, 22, 23.5 and 25V, respectively). The CCD camera as shown in Fig. 18, receiving the projection of laser light on the work-piece, decides the projection direction with respect to the sensor head axis. The angle between the axis of CCD camera and the line from the lens centre to projection spot can be measured. From these two angles, the distance between scanner centre to laser spot and the distance between camera lens and the laser spot are measured. From these information by using law of Triangulation, distance between the work-piece to sensor head can be measured and expressed as Eq. (7).

$$h = \frac{D_x + D_y \tan \theta_2}{\tan \theta_1 - \tan \theta_2} \quad (6)$$

where, h is the measured height by the laser vision sensor, D_x is the distance between the laser spot to the scanner centre, D_y is the distance between the laser spot axis to the axis of CCD camera, θ_1 is the angle of laser light scanner and θ_2 is the angle of direction of laser light projection to the sensor head. By comparing spectrum of welding arc, it has been concluded that optimal diodes selected for arc welding sensors should be of central wavelength 467 nm, 594 nm, 610 nm, 632 nm and 950 nm. Xu et al. [59] proposed a seam tracking system consisting of a rotating lens based light scanning system that produce a circular laser beam and a charge coupled device (CCD) as shown in Fig. 19. The images captured by the vision sensor are examined by an image grabber card.

Based on radius-depth relation of laser cone and camera amplification coefficient, the relation between point of intersection of laser projection $P(X_L, Y_L, Z_L)$ on work piece and corresponding image point in the co-ordinate of sensitivity, $P(X_{pixel}, Y_{pixel})$ has been established. According to the law of imagery and spatial relation, $P(X_L, Y_L, Z_L)$ can be computed from the camera co-ordinate values $p(X_c, Y_c)$ as follows:

Table 2
Papers with utilized methodology and different parameters for vision sensor.

Year	Authors	A	B	C	D	E								
						Wire feed rate	Welding speed	Current	Voltage	Gas flow rate	Frame rate	ROI/Image size	Look ahead Distance	Tracking speed
2002	Kuo and Wu [45]	Prewitt with auto-selecting threshold method, Fuzzy approach., Linear regression method	Steel-2 mm Al-6 mm	–	–	–	–	–	–	–	–	✓	–	–
2004	Ge et al.del. [28].	Diagonal differencing operator, binarizing and statistical analysis, Sliding view algorithm	Stainless steel-4 mm	PAW	–	–	–	–	–	–	–	✓	–	–
2005	Gao and Na [50]	Kalman filter	2 mm	GTAW	–	–	✓	–	–	–	–	–	–	–
2006	Zhou et al.del. [35].	Calibration, Image processing, tracking controller design	Al-alloy	GTAW/Butt joint	–	–	–	–	–	–	–	–	–	✓
2007	Jinqiang et al.del. [34].	Average filtering, masking, Hough transformation, PID controller design	Mild steel-6 mm	GMAW	–	–	✓	–	–	–	–	–	✓	–
2007	Chen et al.del. [33].	Image processing Algorithm.	Al-alloy	microscopic sea	–	–	–	–	–	–	–	–	–	–
2007	Shi et al.del. [48].	Canny operator, shift window	–	Butt joint	–	–	–	–	–	–	–	–	–	–
2007	Shen et al.del. [36].	Image Processing algorithm	–	GTAW	–	–	✓	–	–	–	–	–	–	–
2009	Ma et al.	Auto-regressive model, fuzzy PID seam tracking controller	Al alloy	TIG/ Butt joint	–	–	✓	–	–	–	–	–	–	–
2009	Dinham and Fang [40]	Robot world and hand eye calibration algorithm	–	–	–	–	–	–	–	–	–	–	–	–
2010	Shen et al. [36]	Edge point detection by non-linear square method, PID controller design	Al-alloy-6 mm	AC GTAW/V-groove joint	✓	–	–	–	–	–	–	–	–	–
2010	Ting et al.del. [51].	Dynamic model for mobile robot, Fuzzy and Fuzzy Gaussian Neural Network for designing strategy of coordination control	–	–	–	–	–	–	–	–	–	–	–	–
2011	Micallef et al.del. [42].	K-Cosine detection technique, seam edge reduction method	–	Butt joint	–	–	–	–	–	–	–	–	–	–
2011	Li et al.del. [49].	Fast GPI method	Steel	–	–	–	–	–	–	–	–	–	–	–
2011	Xue et al.del. [44].	Image processing	–	–	–	–	–	–	–	–	–	–	–	–
2011	Chen et al.del. [33].	Image Processing, Controller Design by past mature seam tracking algorithm and Peak current self-adaptive regulating controller	Al-alloy	AC GTAW/V-groove	✓	–	–	–	–	–	–	–	✓	–
2012	Xu et al.del. [30].	Hough transform and Least square method.	–	CO2 gas assisted arc welding	–	–	–	–	–	–	–	–	–	–
2013	Heber et al.del. [41].	Prediction-correction based tracking approach.	–	–	–	–	–	–	–	–	–	✓	–	–
2013	Dinham and Fang [38]	Image matching and triangulation by the utilization of 2D homography.	Steel	–	–	–	–	–	–	–	–	–	–	–
2014	Nele et al.del. [31].	Template matching method	Steel-2.5 mm	GMAW/Butt joint	✓	–	–	–	–	–	–	–	✓	–
2014	Xu et al.del. [32].	Best image selection from intensity gradient, image processing algorithm	Al-3 mm	GTAW/GMAW	✓	–	–	–	–	–	–	–	–	–
2014	Dinham and Fang [47]	Adaptive line growing algorithm.	Al and mild steel	Fillet joint	–	–	–	–	–	–	–	–	✓	–
2015	Xu et al.del. [37].	Improved canny algorithm	Al-3 mm	GTAW and GMAW-Butt joint	✓	–	–	–	–	–	–	–	–	–
2015	Dinham and Fang [39]	Calibration by combining robot kinematics and hand-eye calibration.	–	Butt and fillet joint	–	–	–	–	–	–	–	–	–	–

A- Methodology utilized, B- Work piece material and thickness (mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor parameters



Fig. 16. MotoSight 2D global edition by Motoman [53].

$$x_L - z_L \tan \beta = \frac{-k_x x_c}{\cos \beta} \quad (7)$$

$$y_L = -k_y \cdot y_c \quad (8)$$

where, k_x , k_y are camera amplification coefficients of the position where laser vision sensor has scanned and β is the deflection angle from camera axis to laser axis. Huang et al. [60] introduced an intricate three dimensional eight axis machine tool for laser welding and a vision sensor system called as HUST-SM vision sensor system consisting of a CCD camera and four structured laser light giving integrated 2D and 3D visual data as shown in Fig. 20. The HUST-SM vision system is based up on an integrated approach of triangulation computation method in 3D using red laser light and image measurement method in 2D using green structured light from laser stripe for obtaining all total information about seam. Li et al. [61] proposed a laser vision sensor considering two most important factors i.e. various reflection on work-piece surface due to laser and the position relationship. Structured light triangle measure is the working principle of the designed vision sensor from which the width of weld groove can be computed as shown in Fig. 21.

α is the angle between weldment surface and camera optical axis, is the angle between laser plane and camera optical axis, C is the camera optical axis direction, L is the plane of specular reflection and L is the plane of laser. From these data and width of groove d, the depth of grooved can be measured from Eq. (10) as expresses as follows: where, k_d is the coefficient of proportion of the image co-ordinates. Xiao [62] proposed a laser vision system that can obtain triple range data at a

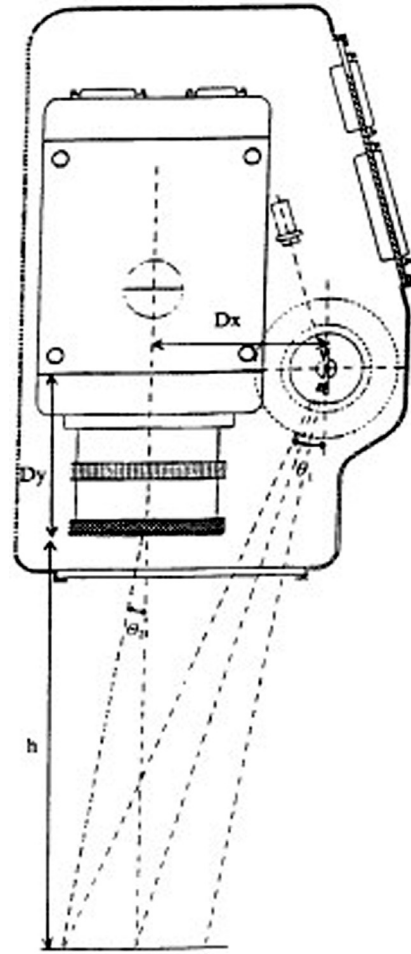


Fig. 18. Laser based vision sensor [58].

time utilizing the three patterns of laser in image. A IIIb- class of laser having 650 nm wavelength and 100 mW power have been utilized for development of TLVS.

The working principle of TLVS is same as the conventional laser sensor based vision system utilizing a projection of triline shaped structural laser on work-piece. Mastui and Goktug [63] introduced an adaptive welding robot system controlled by laser sensor for welding of thin plates with gap variation in single pass.

Fridenfolk and Bolmsjo [64] introduced a seam tracking system in 6D that could decrease the requirement of accurate programming of robot trajectory and geometrical database needed for it. Sung et al. [65] utilized multi laser patterns for developing prototype of multiline laser vision sensor by which multiple range data could be processed

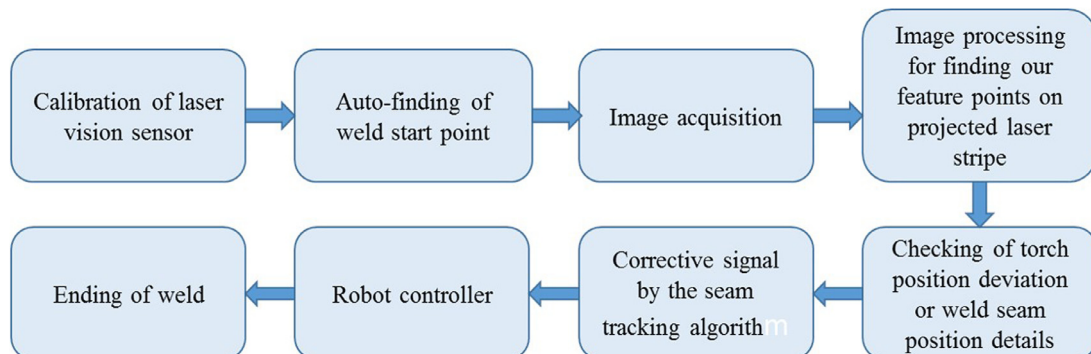


Fig. 17. Method seam tracking with utilization of laser vision sensor.

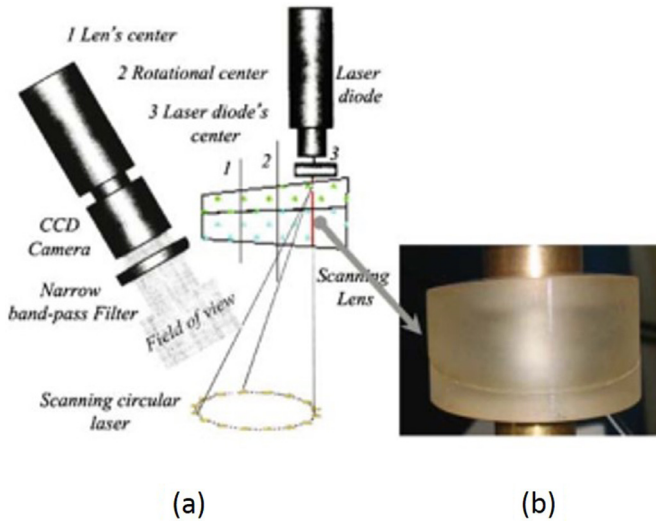


Fig. 19. (a) Vision sensor set up with circular laser (b) scanning lens [59].

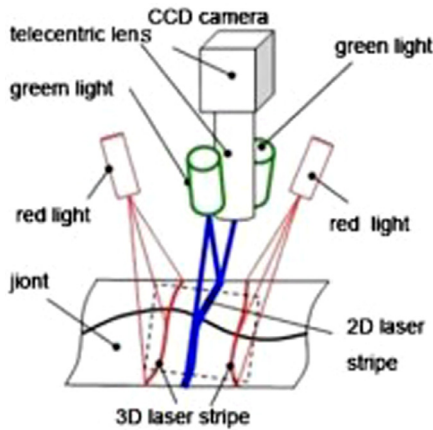


Fig. 20. Configuration of HUST-SM system [60].

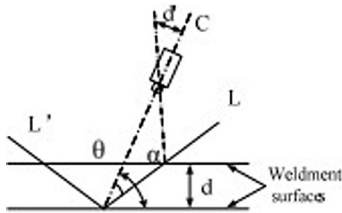


Fig. 21. The principle of triangle measure [61].

simultaneously.

2.3.2. Image acquisition and processing

The image processing generally involve the similar procedure as mentioned for vision sensor in Section 2.2.2. Image processing for a laser based vision sensor for multi-pass welding involve median filtering for noise removal, thresholding by the uniform threshold algorithm to extract point of interest, denoising for removing the spatter, thinning to extract the centre line from the binary image and finally curve fitting [57]. The result of image processing is shown in Fig. 22.

Chang et al. [66] proposed image processing method in which after scanning only filtering is performed before feature point extraction for seam tracking with a mobile robot in ship hull welding application as shown in Fig. 23. Cut off filter has been applied to remove the data that are same or less than average profile depth. Then a Gaussian filter has

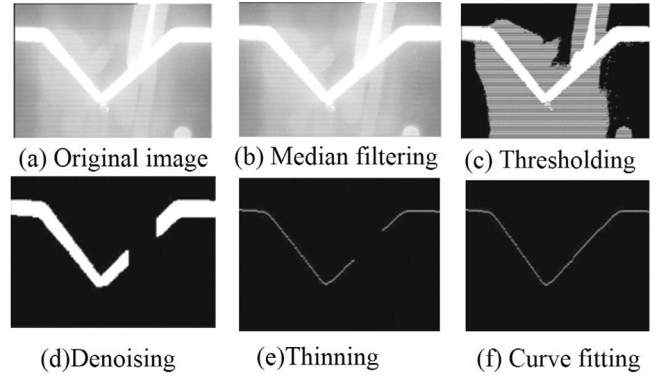


Fig. 22. Result of image processing [57].

been applied to smooth the disturbances of high frequency (Fig. 24).

Wu et al. [67] utilized different filtering techniques to find the optimal filtering technique for removing noise in the image processing procedure for arc welding process. Three filtering technique i.e. Gaussian, Wiener and median technique have been applied for comparison. For checking the efficiency, four statistical algorithms i.e. MSE (Mean Square Error), RMSE (Root Mean Square Error), PSNR (Peak Signal-to-Noise Ratio) and SNR (Signal-to-Noise Ratio) have been used. From the statistical comparison it has been found that MSE and RMSE values for median filter is minimum and PSNR and SNR values for this filter is maximum. This implies that Median filter is the most efficient technique for removing noise under arc environment.

Kim et al. [68] analyzed centre of gravity (COG) method to find the centre co-ordinates from the horizontal and vertical image co-ordinate values and their pixel intensity. Then thresholding is done to filter the noise in the sensing region. Averaging window method is applied to overcome the problems associated with rough surface of centre line having different pixels between consecutive co-ordinates in horizontal direction. Fang et al. [69] described image processing in four steps i.e. pre-processing of image, region of interest calculation and thresholding, skeleton thinning and feature extraction. By projecting the gray value on vertical axis and horizontal axis, the Region of Interest (ROI) can be determined. Then for the segmentation of seam and the laser stripe adaptive threshold values have been computed from projection values. To obtain the skeleton of seam and laser stripe, the fast parallel thinning algorithm has been applied. Then feature lines have been extracted through the use of Hough transformation and least square line fitting can then be utilized for better accuracy of feature lines. Li et al. [70] described the methodology adopted for image processing in two steps: pre-processing and post-processing. At first, the contrast of the image has been enhanced to enhance the contrast of background and target. To remove the noise due to arc, spitting and dust median filtering have been used. Then binarization is applied to extract the target image from the initial image. For getting the mid line more accurately by removal of leftover noises, single point algorithm is then developed. To obtain finer laser centre line inside, stalk transformation has been applied and then slope analysis is done to extract the feature points precisely. Yan et al. [71] described the first stage in image pre-processing as image segmentation and the next stage as the image enhancement in which the goal region has been distributed to two sub-regions. The selection of an appropriate threshold has been done according to the frequency distribution of gray level in the sub-regions. Then Hough transformation has been adopted for detection of the exact line of laser stripe. Li et al. [61] described the image processing for girth seam tracking system. First selection of ROI has been done to remove disturbances in image and to speed up the process. The ROI can be selected by fixing the peaks in intensity projection. The intensity distribution represents the position of laser stripes horizontally. The welding groove profiles were brought out by maximum values of intensity distribution of pixels

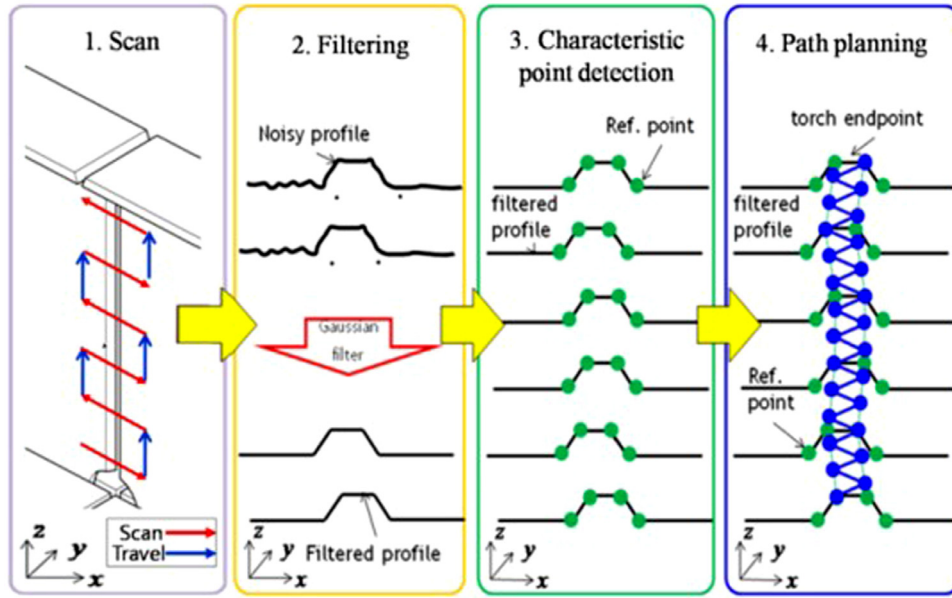


Fig. 23. Flow chart for seam tracking algorithm [66].

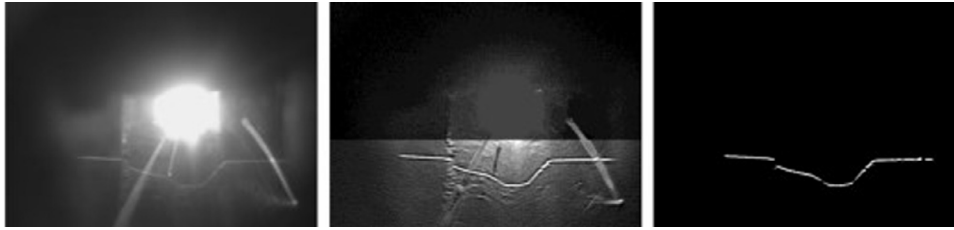


Fig. 24. Result of proposed image processing method [56].

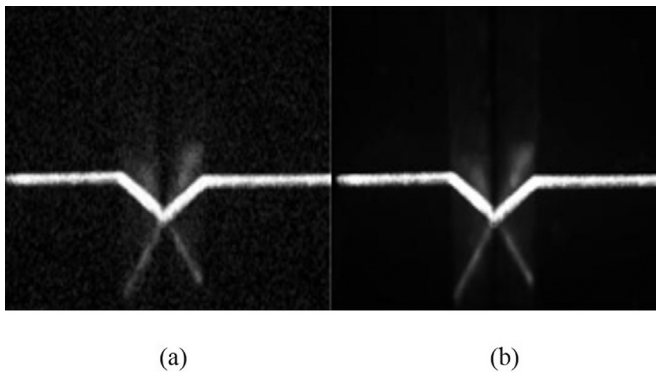


Fig. 25. (a) Image result by Gaussian filtering (b) Image result by Median filtering [74].

based on columns. Xiao [62] proposed an advanced image pre-processing algorithm for the triline laser vision sensor. First median filter has been used to decrease or remove noise because of arc light, reflection and spatter. Then to intensify the laser stripe lines, specialized masks have been utilized. Thinning transformation has been applied to visualize the laser lines more clearly. Finally, the fitting operation is done to obtain positions of seam more accurately. Xiong et al. [72] performed the edge detection by application of Fuzzy technique. The fuzzy rules have been formulated for pixel values with medium negative (MN) and medium positive (MP) in the membership function and output y (white WH or black BL). He et al. [56,73] selected an integrated approach for getting intensity and orientation information so that more precise seam tracking operation can be done and the result is shown in Fig. 25. Gabor filtering has been applied to detect orientation

information of laser stripe. But it has been observed that because of application of Gabor filtering with only one appointed angle, certain part of weld seam profile is not found. To come with a solution for this problem, region segmentation Gabor filtering has been suggested. The outcome of this can be named as combination orientation feature map. The target with large sizes called saliency has to be increased for the laser stripe. Then the dimension metric is computed by application of threshold segmentation and clustering algorithm.

Higher intensity objects like molten pool and arc glare can also be easily identified. As the intensity of laser stripe is not maintaining uniformity, this heterogeneity can also be utilized for increasing its saliency. Lee et al. [74] selected Gaussian filter and median filter to remove components of high frequency, salt and pepper noise respectively. It has been found that by the application of median filter better results can be obtained as shown in Fig. 25. After that, erosion, dilation, opening and closing filter are applied to erase the other noises and to enhance the laser light projection.

Luo and Chen [75] found that the weld start point should be the in weld starting in X co-ordinate, and at weld groove centre in Y co-ordinate and situated above the work-piece by a distance more than arc length. Huissoon [76] designed a calibration system for defining frame of end point of sensor with respect to welding laser frame and robot wrist frame. From this the relationship between co-ordinate frame of robot wrist, welding laser and seam tracking sensor have been established.

2.3.3. Feature point extraction

Gu et al. [57] selected top left and right points, bottom left and right points on the image by the proposed extraction algorithm based on the image pixel values after curve fitting. Chang et al. [66] proposed a differential characteristic-point detection algorithm in which the local

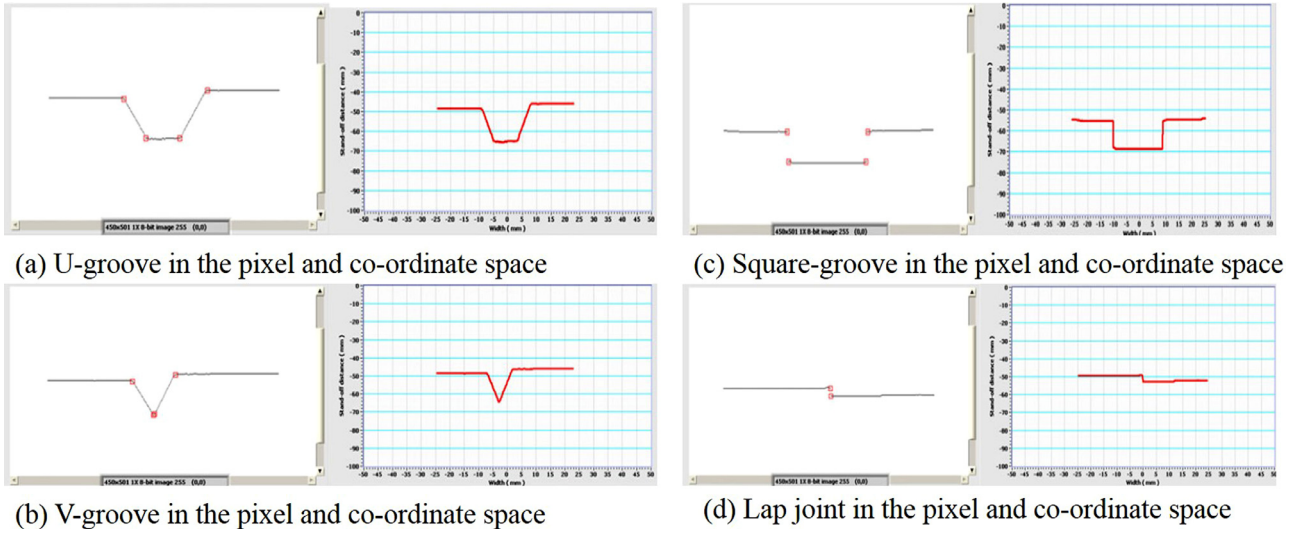


Fig. 26. Feature point detection for different types of joint [77].

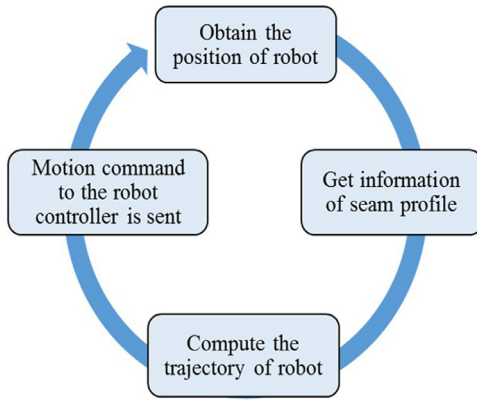


Fig. 27. Flow diagram for robot path planning.

maxima and minima points have to be searched after first and second derivative. After that the final step of path planning is done through linear interpolation of the reference points. Luo and Chen [75,78] proposed a missing point algorithm for the undetected seam point between the previous and next identified seam points based on two-point linear interpolation method. Kim et al. [68] utilized thinning and skeletonization for characteristic point extraction from the laser image. By

applying thinning process centre lines from the regions of image having integer value have been found. For this centre of gravity (COG) method is used efficiently to find the centre co-ordinates from the horizontal and vertical image co-ordinate values and intensity of pixel values. Li et al. [70] applied stalk transformation to obtain finer laser centre line and then slope analysis has been done to extract the feature points precisely for the butt joint with very narrow gap. Yan et al. [71] considered V joint for computing the feature point. The centre point of the weld groove has been regarded as the characteristic point here and the co-ordinates for this are obtained by considering the co-ordinate values of main line and laser stripe. Li et al. [61] detected the central points and the boundary points having lower threshold value in the joint region as the feature of weld seam. Zhang et al. [79] proposed multi-step gray projecting integral method for detection of weld seam. In this method a circular division in image has been applied as the sub-window and a link table has been recognized to store all the edge point values obtained through the gradient image. He et al. [73] applied threshold algorithm and an unsupervised clustering algorithm for obtaining weld seam profile. A double slope plus smoothing algorithm is proposed for extracting feature points for the multi-pass seam. The three feature points are the boundary points representing the swing range of welding torch and the extreme points representing the position of welding torch. Xu et al. [80] proposed an information fusion algorithm for obtaining the edge point position in three dimension from the combined

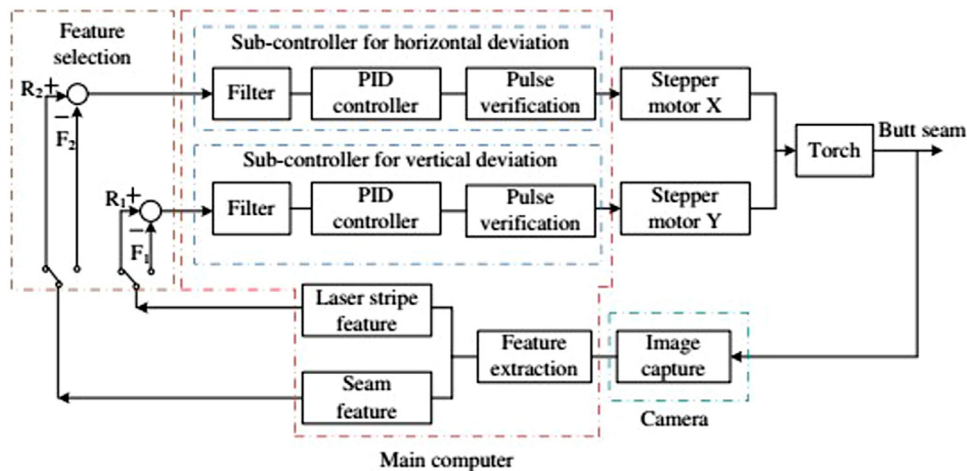


Fig. 28. Block diagram of developed control module [69].

Table 3
Papers with utilized methodology and different parameters for laser sensor.

Year	Author	A	B	C	D	E					Laser wavelength	Tracking speed/ Time		
						Gas flow rate/ Pressure	Welding speed	Wire feed rate	Current	Voltage			Frame rate	Image size
1996	Kim et al.del. [82].	Image processing algorithm through template matching	8 mm	GMA	-	-	✓	-	-	-	-	✓	-	-
1998	Haug and Pritscow [80]	Fuzzy logic based weld groove detection method	-	V-groove joint	-	-	-	-	-	-	-	-	✓	-
1999	Kim et al.del. [81].	Enhanced strip algorithm for extraction of range data	-	Fillet joint	-	-	-	-	-	-	-	-	-	-
1999	Agapiou et al.del. [56].	MIG welding arc spectrum analysis by SPECT spectrograph	-	MIG	-	-	-	-	-	-	-	-	✓	-
2002	Matsui and Goktung [61]	Suggested welding and sensor parameters for different sensors.	-	-	-	-	-	-	-	-	-	-	-	-
2002	Wilson [52]	Analysis about industrial applications	-	Arc welding	-	-	-	-	-	-	-	-	-	-
2002	Pritschow et al.del. [84].	Row and region based filtering, Robert or Sobel for line detection	-	GMA/V-groove joint	-	-	✓	-	✓	-	-	✓	-	-
2002	Huisoon [74]	Calibration system for end point sensor.	-	Laser welding	-	-	-	-	-	-	-	-	-	-
2003	Fridenfolk and Bolmsjo [62]	6D seam tracking system in flexible unified simulation environment	-	GMAW	-	-	-	-	-	-	-	-	-	-
2003	Yoo and Na [83]	Neural network for weld seam feature extraction.	-	-	-	-	-	-	-	-	-	-	-	-
2004	Xu et al.del. [78].	Calibration., Feature point extraction through thinning and Hough transform, visual control module for robot	10 mm	V-groove joint	-	-	-	✓	-	-	-	-	-	-
2004	Luo and Chen [73]	Seam tracking algorithm, auto-calibration,	Titanium alloy	Arc welding/V-groove	-	-	-	-	-	-	-	-	-	-
2005	Luo and Chen [76]	Auto calibration, missing point algorithm to find out seam points.	Titanium alloy	Fillet/lap/Butt joint	-	-	✓	-	-	-	-	-	-	✓
2006	Zach and Vogl [53]	Method of programming of industrial robot through interactive laser projection.	-	-	-	-	-	-	-	-	-	-	-	-
2007	Xu et al.del. [57].	Two-step segmentation technique for weld seam position calculation.	-	GTAW/V /I typr butt joint	-	-	✓	-	✓	-	-	✓	-	✓
2007	Liangyu et al.del. [68].	Image processing, stalk transformation Slope analysis for feature point selection	-	-	-	-	-	-	-	-	-	-	-	-
2007	Yan et al.del. [69].	Image Processing, PID algorithm utilised for control module	Steel-12 mm	SAW/Butt joint	-	-	✓	-	✓	-	✓	-	-	-
2007	Li et al.del. [59].	Feature extraction through Hough transform, image based seam tracking vision control structure	Steel-10 mm	SAW/V-joint	-	-	✓	-	-	-	-	-	-	✓
2008	Kim et al.del. [66].	Centre of gravity method for thinning, Difference of Gaussian for edge detection, Trajectory based control scheme	-	Laser welding/ Butt joint	-	-	✓	-	-	-	-	-	✓	-
2008	Zhang et al.del. [77].	Multi-step gray projecting integral method with circular sub-window for detection of weld seam	Steel	Laser welding/ Butt joint	-	-	-	-	-	-	-	-	-	✓
2009	Sung et al.del. [63].	Multiline laser vision sensor for high speed welding	-	Lap joint	-	-	-	-	-	✓	-	-	-	✓
2010	Fang et al.del. [67].	Image processing, PID controller design based on image based visual control	-	Butt joint	-	-	-	-	-	-	✓	-	-	✓

(continued on next page)

Table 3 (continued)

Year	Author	A	B	C	D	E				Laser wavelength	Tracking speed/ Time			
						Gas flow rate/ Pressure	Welding speed	Wire feed rate	Current			Voltage	Frame rate	Image size
2011	Xiao [60]	Image processing for trillines laser vision sensor	–	V-joint	–	–	–	–	–	✓	–	–	✓	–
2011	Xiong et al.del. [70].	Fuzzy logic for edge detection method	–	–	–	–	–	–	–	–	–	–	✓	–
2011	Koch et al.del. [85].	Look ahead visual control scheme for unknown three dimensional path	–	–	–	–	–	–	–	–	–	–	–	–
2012	Huang and Kovacevic [75]	Image processing algorithm and corner points detection algorithm	–	–	–	–	–	–	–	✓	–	✓	–	–
2012	Chang et al.del. [64].	Characteristic point detection point algorithm	Steel	V-joint	–	–	–	–	–	–	–	–	–	–
2013	Huang et al.del. [58].	HUST-SM vision sensor system	–	–	–	–	–	–	–	–	–	–	–	–
2013	Gu et al.del. [55].	Image processing, Fuzzy-P controller	–	MAG/V-joint	✓	–	–	✓	–	–	–	✓	–	–
2014	Wu et al.del. [65].	Optimal filtering technique by statistical techniques	SM490 steel	GMA	–	–	–	–	–	–	–	–	–	–
2014	Lee et al.del. [72].	Optimal image processing algorithm	SM490Steel-9.5 mm	GMA/V-joint	–	–	–	–	–	–	✓	–	–	–
2015	Quan and Bi [79]	Mathematical model for analysis and calculation of variation in width and position of assembly gap.	2 mm	CNC machine controlled welding	–	✓	–	–	–	–	–	–	–	–
2015	He et al.del. [54].	Saliency based visual attention model	Low carbon steel-30 mm	MAG/V-joint	–	–	–	✓	–	–	✓	–	✓	–
2016	He et al.del. [71].	Visual attention model, polynomial fitting plus derivatives methodology.	Low carbon steel-30 mm	MAG/V-joint	–	–	–	–	–	–	✓	–	–	–

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor Parameters



Fig. 29. TH6D laser sensor from Scansonic [88].

parameters of laser plane and camera. He et al. [56] proposed a three step technique for extracting the profile of weld seam. The steps involved are threshold segmentation, execution of Euclidean distance based clustering algorithm and finally removal of small sized clusters. Huang and Kovacevic [77] developed an algorithm based on second central difference of row index of every pixel detected on laser stripe. The result of the proposed algorithm for different joints are represented in Fig. 26. Quan et al. [81] proposed a mathematical model for transformation between features of image and assembly gap being welded. The real world co-ordinates values of trajectory path have been calculated theoretically and then compared with the values obtained from the image of projected laser stripe. The accuracy of the system has been found to be suitable for high quality welding and adjusting the weld parameter simultaneously.

Haugand and Pritscow [82] developed a feature based fuzzy algorithm for distorted height profiles of V groove. The features calculated after line segmentation like segment length, gradient and distance to important profile part are considered as the input parameters for the fuzzy algorithm. The co-ordinate values of line segments and degrees of membership for welding groove edge positions are then calculated. The mean value of the edge positions has been considered as the weld groove position. Kim et al. [83] developed a geometric model for determining torch position, orientation and robot path generation simultaneously by integration of information obtained by laser vision sensor range data and CAD data. An enhanced strip algorithm has been used for extraction of range data by laser vision sensor. Kim et al. [84] proposed segmentation of joint profile and template matching algorithm to obtain the features of joint by defining the matching cost in terms of measures of straightness, angle and length. Yoo and Na [85] utilized neural network technique for extraction of weld seam feature points. The sum of each image pixel co-ordinates and the maximum difference of two consecutive pixel co-ordinates have been considered as the input parameters. Two hidden layers, each having 10 nodes have been developed to get type of pattern as output. Finally line segmentation method is applied to extract the feature points from the image patterns. Lee et al. [74] utilized skeleton and contour on average method for extraction of centre line. Then further Harris algorithm with slope detection algorithm has been explored for feature point extraction. Pritschow et al. [86] proposed row based processing, region based processing and line detection filtering for the detection of groove

Table 4
Papers with utilized methodology and different parameters for ultrasonic sensor.

Year	Authors	A	B	C	D	E			F	
						Wire feed rate	Welding Speed	Current		Voltage
1984	Estochen and Neuman [86]	Algorithm for performing surface height sampling in 2D	–	Arc welding	–	–	–	✓	–	Robotic sealing, flame cutting
1989	Maqueira et al.del. [88].	Trajectory generating algorithm for the torch position	Al-3.175 mm	V-joint	–	–	–	✓	–	–
1994	Bastos et al.del. [87].	Model based on echo parameter over flight time	–	Butt/Lap joint	–	–	–	✓	✓	–
1998	Zhang et al.del. [89].	Optimization model for obtaining a wide band width, Receiving circuit design	steel/mild steel-12 mm	GTAW/GMAW-V-joint	–	–	–	✓	–	Steel jar welding

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor parameters, F- Applications

Table 5
Papers with utilized methodology and different parameters for electromagnetic sensors.

Year	Authors	A	B	C	D				E		F
					Wire feed rate	Tip to work piece distance	Current	Voltage	Sampling time	Look ahead distance	
2004	Chung et al.del. [106].	Data regression algorithm, moving average method	–	GMAW/Butt joint	–	✓	✓	✓	✓	✓	–

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor parameters, F- Applications

geometry. From experiments it has been found that line detection filtering is the most efficient way of reducing the noise and determination of groove dimensions.

2.3.4. Controller design

The features of weld seam extracted by the image processing technique are then used by the controller module of seam tracking system to guide the robot along the weld seam path. Gu et al. [57] developed a seam tracking controller consists of two types of controllers i.e. the fuzzy controller for small offset region and the proportion controller for large offset region. Depending on the data of accurate extraction of weld deviation i.e. difference of features of reference image and feedback image, the Fuzzy-P controller has been used to control the torch to travel along joint with constant velocity. Luo and Chen [78] developed a seam welding path conversion algorithm to translate the seam path to actual welding path considering the actual start position and travel of torch in Y and Z axis with the look ahead distance (LAD). A delay time or look ahead time has been considered for solving the problem associated with the algorithm for LAD. Then the deviation for seam point in Y and Z axis has been calculated by which the accumulated error can be computed. Finally, according to the accumulated error the torch is shifted by an equivalent amount. Kim et al. [68] developed trajectory based control in which the laser stripe sensor led the robot to the seam path without giving prior information of robot path as shown in Fig. 27.

The welding direction has been computed by predicting the position of robot from average velocity of robot and current position step of robot. Then linear interpolation has been used to identify the location of seam that is not presented in seam location buffer. Fang et al. [69] designed two sub-controllers to adjust torch deviation in horizontal and vertical direction as shown in Fig. 28. To achieve this the features of reference image have been calculated before the beginning of welding process while aligning the torch with seam manually. At every sample time, the feedback images are sensed and features are extracted to calculate the deviations of torch. Error filtering according to Gaussian function has been done for removing rapid changes in errors. Incremental PID controller has been utilized as sub-controller which add the scaling factor to controller according to working the feature of stepper motor.

Yan et al. [71] developed a control module that used the error between the reference image and the captured image in horizontal and vertical direction as input for PID algorithm to compute the number of pulses of the stepper motor. This would be used to control the cross slide and ultimately orient the torch in corrected welding path. Xu et al. [80] proposed image based and position based robot visual control for removing tracking error and for determining the direction or position setting of robot respectively. After calculation of feature point position in 3D from the images, the master computer gave results in terms of parameters of robot line movement. The selection of control solution, resolution for reverse kinematics and motion planning have been done through slave computer. In slave computer, the values of position setting have been shared by each joint controller indicating speed closed-loop and position closed-loop control. Koch et al. [87] presented a contour following algorithm for removing optical noises, alignment of torch along unspecified shape of seam joint with very less tracking error

while controlling the robot velocity in real time. Fridenfolk and Bolmsjo [64] proposed a control system based on position, trajectory tangent vector and pitch control. The proposed control system has output of next position of tool centre point from the input of current position of tool centre point. The summary of papers studied for laser vision sensors are listed in Table 3.

2.3.5. Laser vision sensor available in market

The TH6D laser vision sensor available from company Scansonic is a non-contact type triangulation sensor in which multiple laser beams are used to detect the seam position, weld gap and joint edge offset. These data are transmitted to robot controller for positioning the tool accurately with respect to weld surface [88]. The laser vision sensors are also available from other companies like MOTOSense from YASKAWA, DIGI-LAS/MDL from SERVO-ROBOT, Smart Laser Pilot(SLPi) from Meta Vision Systems and scanCONTROL from MICRO-EPSILON etc. The image of TH6D sensor is shown in Fig. 29

2.4. Ultrasonic sensor for weld seam tracking

For the first time, Estochen [89] gave preliminary idea about utilization of ultrasonic sensing in seam tracking. The sensor works on the principle of relation of ultrasound propagation interval in air to the source of ultrasound, point of reflection distance, reflection point and receiver distance. Beodiano [90] used ultrasonic sensor with a 220 kHz piezoelectric transducer and proposed periodic sampling method of echo amplitude for detecting the seam geometry. Maqueira et al. [91] developed an ultrasonic sensor based automatic seam tracking system for getting two dimensional geometric information about the joint in real time. The sensor basically consists of a transducer driven by LAM80/8 at 100 kHz. The strategy for transducer positioning with simultaneous measurement of transit time and echo pulse amplitude by the proposed torch sensor arrangement helps in removing confusions obtained from symmetry in acoustic signals and other environmental factors. The periodic voltage pulses through the pressure wave (generated through air by the voltage pulse applied to transducer) are reflected by edge of the joints. In the system, then a peak detector has been used to measure the occurrence of maximum amplitude. Finally, the proposed trajectory-generating algorithm has been applied by the micro-computer to obtain the information for on-line tracking. Zhang et al. [92] optimized the transmission efficiency by matching impedance for enhancing the accuracy of weld seam measurement of high frequency ultrasound based seam tracking system. It has also been observed that a high resolution of ultrasound beam can be achieved by proper designing of transducer focal length. The summary of papers studied for Ultrasonic sensors are listed in Table 4.

2.5. Electromagnetic sensor for seam tracking

You and Kim [93] designed an electromagnetic sensor to detect the weld seam of a butt type weld joint by utilization of change of inductance of coil through the production of eddy current. The sensor used is actually a coil inside a bridge circuit to which alternating current is supplied. When the sensor is moved towards work-piece material, according to Lenz law eddy current and magnetic flux are induced

Table 6
Papers with utilized methodology and different parameters for Infra-red sensing.

Year	Authors	A	B	C	D	E				
						Welding speed	Tip to work piece distance	Current	Voltage	Gas flow rate
1992	Nagarajan et al. del. [92].	Effect of weld parameters on plate surface temperature.	AISI 1008 steel-6.35 mm	GTAW	✓	✓	✓	✓	✓	✓
1995	Banerjee et al. del. [93].	Microscopic temperature gradient analysis	–	GTAW/GMAW	✓	✓	–	✓	✓	✓
2001	Wikle et al. del. [94].	Monitored the variation of temperature distribution	carbon steel-3/8 inch	GTAW/SAW	–	–	–	–	–	–
2012	Gao et al. del. [91].	Analysis of gray value gradient and thermal gradient parameter	stainless steel-10 mm	Laser welding/Butt joint	✓	✓	–	–	–	✓

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor parameters

on metal surface resulting in change of inductance of coil. The magnitude of change is a measure of distance between sensor and work-piece surface and other imperfections. Data regression algorithm has been applied for generation of weld line. The accuracy of tracking is improved by application of moving average method. The summary of papers studied for electromagnetic sensors are listed in Table 5.

2.6. Infrared sensing for weld seam tracking

Gao et al. [94] proposed an infrared image sensing with visual sensing technique that basically consists of an infrared sensitive high speed camera oriented in off axis of laser welding head. The deviation between the centre of weld seam and focal point laser beam has been determined from thermal gradient parameter derived from gray value gradient of infrared image and key hole margin of molten pool image. Nagarjan et al. [95] utilized a thermal imaging system to determine variations in weld bead width, depth of penetration and torch offset in arc welding. By an infrared camera, the infrared radiation representing the thermal distribution of weld plates has been detected. Inflections technique and absolute temperature technique are applied then to measure the weld bead width. By analyzing the data of temperature distribution, the depth of penetration of joint has been detected. The asymmetry in the distribution of surface temperature detects the torch offset from which the orientation of the torch has been obtained. Banerjee et al. [96] proposed a gradient method for real time control weld seam in by monitoring geometry of weld bead considering parameters like alteration in plate thickness, presence of other minor elements, composition of shielding gases in TIG welding. Also the weld quality has been controlled through monitoring weld bead depth and width. Wikle et al. [97] first developed a heat transfer model to predict the optimal position of infrared sensor point with respect to welding arc and the temperature distribution of welding operation. Then controller system has been designed to control the penetration depth, weld bead dimension and seam tracking with the infrared sensor. Finally the wire feed rate and welding voltage have been varied with respect to output signal of controller according to variation in plate dimensions. The summary of papers studied for electromagnetic sensors are listed in Table 6.

2.7. Composite sensor for weld seam tracking

Xu et al. [98] developed a composite sensor consisting of arc sensor and visual sensor for obtaining the information of weld seam in three dimension. An improved processing algorithm has been proposed to acquire features values from the weld images and arc voltage signals. A relational model for establishing relationship between arc voltage and arc length has been obtained. From the image processing, the deflection from pool centre point and seam centre line is computed in real time. The summary of papers studied for electromagnetic sensors are listed in Table 7.

2.8. Tactile sensor for weld seam tracking

Presern and Gyergyek [99] developed a tactile sensor with one finger and two degrees of freedom for finding and tracking of three dimensional trajectory of seam in robotic arc welding. The tactile signal has been used for identification of feature locations on the object surface and then detail touching of the featured locations have been performed for obtaining the database of the sensing system. A micro-computer is used for organizing and reducing data of sensor in a form that could be matched to a knowledge network. For this, a hierarchical organization of database of object has been proposed and further a matching and search algorithm have been utilized. Suwanrachatananee et al. [100] presented a tactile sensor with welding torch framework for robot controller and a dynamic detecting system to figure out a three dimensional edge tracking system. The tactile sensor

Table 7
Papers with utilized methodology and different parameters for composite sensors.

Year	Authors	A	B	C	D						E
					Welding speed	Wire feed rate	Current	Voltage	Gas flow rate	Duty cycle	Tracking time
2012	Xu et al.del. [95].	Based on technology of visual sensor and arc sensor	Al alloy-3 mm	GTAW	✓	✓	✓	✓	✓	✓	✓

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor parameters,

Table 8
Papers with utilized methodology and different parameters for tactile sensor.

Year	Authors	A	B	C	D			E			F
					Welding speed	Current	Voltage	Tracking speed	Look ahead distance	Sensing Force	
1983	Presern and Gyergyek [96]	Beam search algorithm	–	–	–	–	–	–	✓	–	–
2008	Suwanratc-hatamanee et al.del.[97].	Tactile sensor with two swap functions	–	–	–	–	–	✓	–	–	Welding and Inspection
2009	Suwanratc-hatamanee et al.del. [98].	Techniques for object sensing and object normal sensing	–	–	–	–	–	–	–	✓	Welding, humanoid robot feet

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor parameters, F- Applications



Fig. 30. Tactile SEAM TRACKER from LINCOLN.Electric [102].

actualized on the robot hand comprised of two programmed swap functions. One of them is tactile sensing utilized for detection of weld edge which comprised three force sensitive resistors organized in triangular manner. The other one is simulating MIG welding torch associated with moving technique. Suwanrachatananee et al. [101] further utilized tactile sensing device and its control mechanism to identify shape of the object, interaction of human-robot in industrial applications. From experiments it has been found that the proposed sensing system can be utilized for obtaining object data in real time typically for tracing of weld trajectory applications. The summary of papers studied for tactile sensors are listed in Table 8.



Fig. 31. Tactile SEAM TRACKER from LINCOLN.Electric [103].

2.8.1. Tactile sensor available in market

The tactile SEAM TRACKER from LINCOLN.Electric company uses a tip depending the type and size of seam which is dragged in the seam to detect the seam position. When there is a little change in the position of seam, it moves the tip of the sensor and the response is received by the tracking system to move the cross slides and the torch to align it along the correct position [102]. The image of tactile SEAM TRACKER from LINCOLN.Electric company is shown in Fig. 30

2.9. Touch sensor for seam tracking

This is one of most simple but not so versatile sensor which tracks the seam by wire sensing. No other hardware is used for the tracking; all the process is performed via the weld wire. The touch sensor FD-WD from OTC, Daihen Inc is available in market and the image is shown in Fig. 31 [103].

2.10. Weld pool observation sensing for seam tracking

Pietrzak and Packer [104] used vision sensor for maintaining weld penetration by establishing relationship between the weld pool geometry with weld penetration. A machine vision algorithm has been proposed to find the weld pool edges and calculation of weld pool width irrespective of presence of heat affected zone, impurities of weld pool,

Table 9
Papers with utilized methodology and different parameters for weld pool observation sensing.

Year	Authors	A	B	C	D	E				Tracking time	
						Wire feed rate	Welding speed	Current	Voltage		Gas flow rate
1994	Pietzak and Packer [99]	Mission vision algorithm, PID algorithm	Inconel-2.54 mm	GTAW	✓	✓	✓	✓	✓	✓	–
1995	Kovacevic et al.del. [100].	Image processing algorithm, current based adaptive control algorithm	Stainless steel-1 mm	GTAW	–	✓	–	✓	–	–	–
1997	Kovacevic and Zhang [101]	Real time image processing of free weld pool surface.	–	GTAW/GMAW	–	–	–	–	–	–	✓
2007	Shen et al.del. [102].	Vision based image processing method and weld penetration.	Al alloy-6 mm	GTAW-Y groove	✓	✓	–	–	✓	–	–
2008	Pal et al.del. [103].	Artificial neural network, multiple regression analysis	Mild steel-8 mm	PMAW/V-groove	–	✓	–	✓	–	–	–
2009	Chen and Wu [104]	Image processing algorithm, ANN model for dynamic features of weld pool top side.	Low carbon steel /Al-2 mm	GTAW	–	–	–	✓	–	–	✓
2010	Ma et al.del. [105].	Measurement of geometric features of weld pool from the gray level distribution features.	LF6 Al-2 mm	pulsed GTAW/Butt joint	✓	✓	✓	✓	✓	–	✓

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor parameters

presence of scratches, marks etc. A PID control algorithm has been developed to maintain the width of weld pool by varying the weld current. Kovacevic et al. [105] utilized high shutter speed camera with illumination of pulsed laser to sense weld features. Based on the image characteristics, the weld pool edges are detected. A control algorithm has also been proposed with an online identifier. Kovacevic and Zhang [106] utilized a CCD camera with pulsed laser projection on weld pool surface for getting clear image about laser stripe, heat affecting zone edges and boundary of weld pool. The reflected stripe of laser provided critical information about weld pool surface. The reflection pattern can be further used for controlling weld penetration. Shen et al. [107] proposed a vision based image processing algorithm to compute the weld seam gap in real time. For the process control, two methods have been utilized i.e. PID method for adjusting wire feed rate and Fuzzy method for current for helping the robot in selection of proper welding process with weld gap variation. Pal et al. [108] first used response surface methodology for designing the welding experiments. Then artificial neural network has been used for developing multi-layer neural network model. The RMS value of welding current and voltage measured from arc sensor have been considered as input parameters and ultimate tensile strength of weld plate has been considered as the output parameter for the developed model. Multiple regression analysis then further considered comparison with the developed model. Chen and Wu [109] developed image processing algorithm for a vision sensor consists of Gauss filtering, edge points regressing for determining maximum width and measurement of top side half-length of weld pool. Then dynamic characteristics of weld pool have been modeled on the basis of controller design. For this back-propagation type artificial neural network has been selected considering peak current, pulse duty ratio as input variables and maximum width and half-length as output parameters. Ma et al. [110] developed a binocular vision system consists of two CCD cameras based on two-level reflection mechanism. From the gray level distribution of region of interception seam gap positions, centre line and gap width have been obtained. The summary of papers studied for weld pool observation are listed in Table 9.

3. Other approaches for seam tracking

The other approaches may not use any sensor for weld seam position extraction. The mobile robot mainly use sliding mode control for seam tracking. While CAD based model of weld seam path can also be incorporated to find out the robot position with respect to actual weld seam co-ordinates.

3.1. Mobile robot with seam tracking system

Chung et al. [111] proposed a nonlinear controller based sliding mode control for a welding mobile robot of two wheels for seam tracking through a smooth curved joint path at a constant welding speed. The dynamic model has been developed in terms of Cartesian co-ordinates of the mobile robot. Definition of tracking errors and the determination of sliding surfaces are considered as the preliminary requirements of the controller so that the error value would be zero. The errors have been measured simply by two potentiometers by taking both controllable and fixed torch. Liu et al. [112] proposed an adaptive sliding mode control algorithm without using the dynamic models of robot for the seam tracking application. Stability analysis of the proposed algorithm has been performed with the Lyapunov function. From the analysis, it has been found that the proposed algorithm has better performance in tracking as both position tracking error and position velocity tracking converged to zero with a faster convergence speed. The summary of papers studied for mobile robot sensing are listed in Table 10.

Table 10
Papers with utilized methodology and different parameters for mobile robot sensing techniques.

Year	Authors	A	B	C	D				E	F
					Wire feed rate	Welding speed	Current	Voltage	Sampling time	
2004	Chung et al.del. [106].	Sliding mode controller design for fixed torch and controllable torch	–	–	–	√	–	–	√	–
2015	Liu et al.del. [107].	Sliding mode control algorithm verified by Lyapunov stability theory	–	–	–	–	–	–	–	Large scale structure welding

A- Methodology utilized, B- Work piece thickness(mm), C- Type of welding/weld joint, D- Weld parameters, E- Sensor parameters, F- Applications

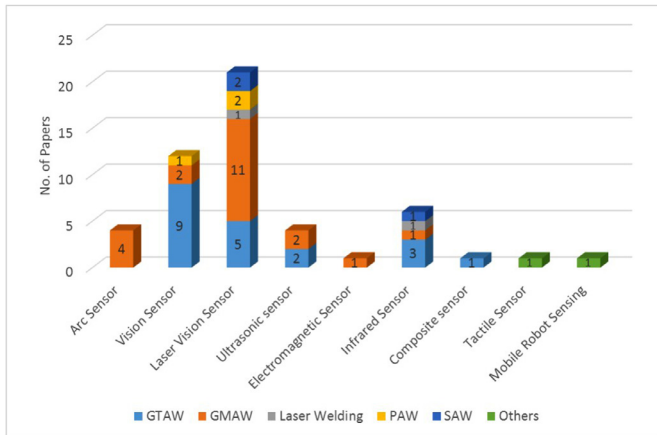


Fig. 32. Proportion of different sensors used for different types of welding in literature selected for review.

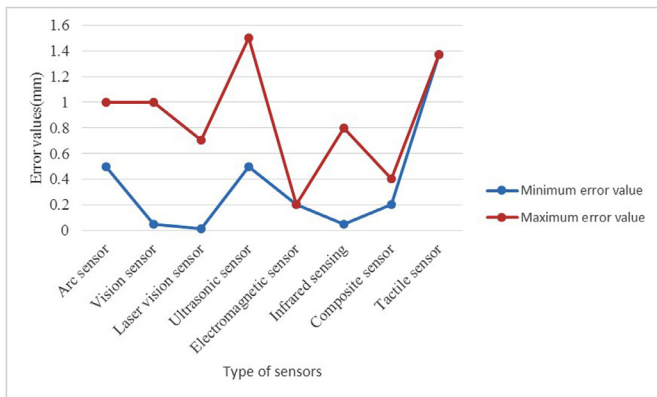


Fig. 33. Accuracy of different sensors utilized for review.

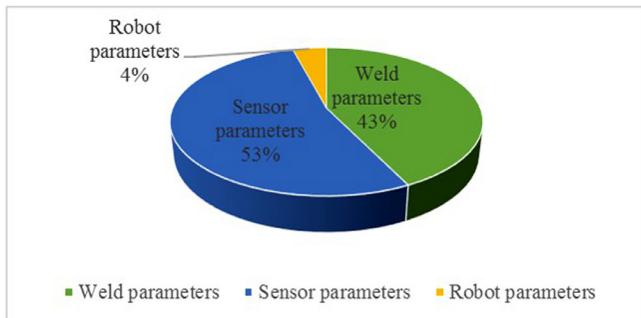


Fig. 34. Proportion of different parameters specified in papers.

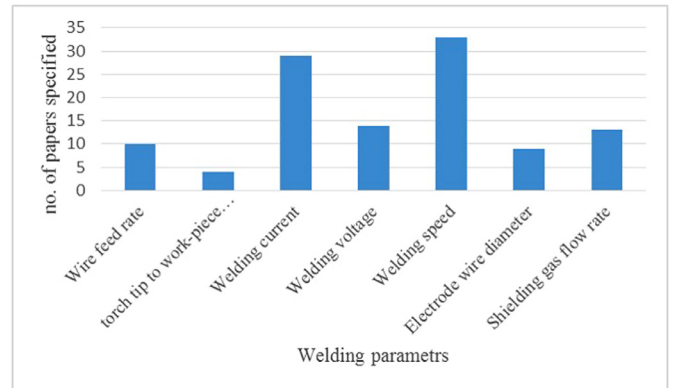


Fig. 35. Number of weld parameters specified.

3.2. CAD based seam tracking

CAD based models are generally used to develop off line programs for robot functioning. Deepak et al. [113] performed seam tracking operation in Computer Aided Design (CAD) environment with the application of a new sewing technique. For this, a 3-axis robot manipulator has been developed in virtual environment for doing welding operation. To guide the welding torch along the weld seam line, forward and inverse kinematics models have been developed for the robotic arm. Neto et al. [114] developed a novel CAD based model for robot programming based on CAD drawing in three dimension in Autodesk Inventor platform. The CAD drawing refers to the trajectory path to be followed by the robot end effector in random environment. The proposed robot programming system has been applied for seam tracking tasks for adjustment of end effector position in the required weld seam path. Carvalho et al. [115] discussed about application of off line programming in terms of 3D CAD model in flexible welding manufacturing cells. The advantages of offline programming for tool movement in different control level have been discussed in detail over on line method.

4. Discussion

Different sensors and their techniques for seam tracking in robotic welding have been discussed in this paper. The proportion use of different sensors for seam tracking for different types of welding is described in Fig. 32.

It can be observed from the figure that for seam tracking operation the maximum used sensor in industries and research works now- a-days is laser sensor and mostly for arc welding. The vision sensor is the next most used sensor for seam tracking operation in industries. The reason may be the accuracy and ease of use with laser sensor cannot be obtained by other sensors. It is also observed from the figure that sensors are used in maximum range for robotic arc welding like TIG, MIG and PAW. The seam tracking error obtained by different sensors have been analyzed from literature survey. The maximum and minimum error

Table 11
Advantages and disadvantages of different sensors.

Sensors	Merits	Demerits
Arc sensor	✓ Cheap compared to vision based sensors used for seam tracking. ✓ No involvement of complex processing for seam tracking like image processing in vision sensor	✓ Seam tracking operation has been done only in real time by finding only the deviations in torch position. Other joint and weld parameters like gap size, weld bead size, robot jerking for rough joint edges or sudden changes in joint paths could not be identified by the proposed models. ✓ Limited use to only MIG welding. ✓ Generally utilized for torch's weaving motion. ✓ Start point and end point for welding cannot be defined accurately by the proposed algorithms. ✓ Cannot be utilized for sheet metals or metal of thickness lower than a particular critical value ✓ Need of extra calculations for camera calibration purpose.
Vision sensor	✓ Can be utilized for both detecting weld seam position and weld groove dimensions ✓ The weld pool images can also be taken during welding for weld parameter control ✓ Easy installation and easy availability. ✓ Cost effective process.	✓ Precautions to be taken during real time seam tracking for lens protection. ✓ Two dimensional data can be obtained with vision sensor.
Laser vision sensor	✓ The error values in seam tracking process is minimum ✓ The image processing method is much simpler as the weld features can be directly extracted from the image of the projected laser stripe on the weld seam. ✓ Weld seam position and groove dimension in three dimension can be obtained simultaneously with very less deviation. ✓ Weld parameters can be controlled in real time. ✓ Can be utilized for wide range of welding applications compared to other. ✓ Can be utilized for wide range work-piece material and thickness such as steel, Al, titanium etc. ✓ Weld quality inspection after welding can also be done by measuring the weld bead shape and size.	✓ High cost ✓ Installation procedure of sensor is not simple.
Ultrasonic sensor	✓ Any presence obstacle for work-piece can also be explored. ✓ Weld quality can also be inspected	✓ High cost ✓ One major parameter of sensor i.e. speed of sound is dependent on temperature, so difficult to model seam tracking in real time ✓ The sight line should not deviate from normal to surface. ✓ Generally utilized for thin sheet metals. ✓ Groove dimensions cannot be obtained
Electromagnetic sensor	✓ Not affected by arc light, fumes. ✓ Can be easily utilized for any work-piece surface condition whether with paint or scratch.	✓ Generally utilized for Gas tungsten welding ✓ Infrared cameras are of high cost and fragile.
Infrared sensing	✓ Temperature distribution around weld pool can be analysed. ✓ Weld quality can be monitored in real time. ✓ Plate surface conditions can be examined.	✓ Installation of sensor is quite complex ✓ Less accuracy. ✓ Can be used only for offline seam tracking. ✓ Joint dimension cannot be obtained. ✓ Very less accurate
Composite sensor	✓ Advantages of both arc sensor and vision sensor can be obtained.	
Tactile sensor	✓ Low cost compared to other ✓ The weld position in three dimension can be obtained.	
Touch sensing	✓ The wire tip or torch of welding sensor can be utilized for weld seam sensing. ✓ The offset of work-piece can be recognized	
Weld pool feature observation	✓ Weld pool characteristics and geometry can be observed on the spot. ✓ Weld quality can be controlled through weld parameter variation.	✓ Weld seam position cannot be obtained. ✓ Design of control module is not easy. ✓ The image processing method should be able to give a clear understanding of image

values of different sensors are shown in Fig. 33.

It has been observed that the accuracy achieved by laser vision sensor is better compared to other sensing techniques as the minimum error with this sensor is as low as 0.01 mm. After that the most utilized and accurate sensors are vision sensors. But infrared sensors may be accurate but is less utilized for seam tracking operation of robotic welding.

In the study, different papers have considered different specifications for weld parameters, sensor parameters and robot parameters. The proportion of paper specified the above mentioned parameters are described by Fig. 34.

From the figure it can be observed that the weld parameters and sensor parameters are specified properly and taken into consideration for in process control of weld quality but robot parameters are not explored effectively by most of the authors. The weld parameters specified mainly include wire feed rate, welding voltage, welding current,

torch to work-piece distance, type of electrode and shielding gas, gas flow rate, welding speed, torch tip to work-piece distance, weld frequency, power output etc. Similarly the sensor parameters specified for vision sensor and laser vision sensor are frame rate, image size, look ahead distance, region of interest, seam tracking speed, detection range, sampling rate, scan per mm, type of weld path etc. In robot parameter only robot travel speed is specified by some authors. The different weld parameters and their proportion of specification in different journals is described in Fig. 35.

It has been observed that welding current, welding speed and the welding voltage are the maximum specified parameters in different papers. These parameters can be considered as the most effective parameters in controlling the weld quality during the welding process. The welding speed plays an important role in controlling the robot travel along the correct weld seam path. This parameter has been considered in controller design for robot motion planning with the

sensory feedback.

The merits and demerits of different sensors are described in Table 11.

5. Conclusion and future scope

From the review analysis, following conclusions are identified.

- The basic purpose of sensors in robotic welding for seam tracking is to automate the industrial robots in teach and playback mode so that the welding process can be performed according to the sensed weld seam path.
- The different sensors, their working principle and development of controller for robot path planning has been discussed.
- It has been observed that soft computing techniques like neural network and fuzzy approach are generally used for controller design for weld parameter adjustment during the welding process. In some papers, for robot path planning proportional integral derivative (PID) controller has also been used with input as the error values between reference position data and sensed position data.
- Among different sensors the laser vision sensors and vision sensors are the mostly used sensors for seam tracking task. These sensors can be utilized for obtaining weld seam path positions as well as weld seam geometries for different types of weld seam joint and weld path with varying weld gap. Moreover, these sensors may also be used for different types of work-piece materials with different thickness.
- In pipe line welding, ship building applications involving steel work-pieces material arc sensor has been used to perform MIG welding. The arc sensor generally used for thin metal plate welding having thickness less than 0.5 mm.
- Vision sensors are used in plasma, TIG and MIG arc welding to weld steel pipe lines, steel and aluminium flanges. The thickness of work material may vary from 2 mm to 10 mm with various type of weld joints like butt joint, fillet joint and V-grooved joints with gap or without gap.
- Whereas, laser sensor has been utilized in wide range of applications like welding double hulled shipyards, marine pressure vessel, pipe welding, oil tank welding, sewing, grinding machines, assembly gap welding, automobile sector and ship building etc. The work-piece material in these applications include various type of steel like mild steel, low carbon steel, stainless steel etc., aluminium and titanium based alloys having highly reflective surface with various type of welding like, TIG, MIG, submerged arc welding, plasma arc welding, CNC machine controlled welding and laser welding. The thickness of the material may vary from 2 mm to 30 mm with very large length as the use of sensor include both single pass and multi-pass welding. The types of joint include V-type groove joint with varying groove angle, root gap, lap joint, fillet joint, butt joint and U shaped step joint. The type of robot utilized with laser sensor may be industrial robot with 2–6 axis, mobile platform manipulator or bogie type robot etc.
- The use of ultrasonic sensor has been found in applications like part location, robotic gluing, sealing, flame cutting and steel jar welding with very less range of use for seam tracking applications. The type work-piece material may be steel and aluminium with thickness in the range of 10–12 mm for mainly TIG and MIG welding. The mobile robot sensing are generally used for large scale structure welding and inspection. Whereas, other sensors have very limited use in industries or not specified properly about their typical applications.

5.1. Future scope

In the future, the following research aspects may be considered in the fields of robotic welding:

- Though different weld parameters and sensors parameters have been specified in different papers but robot parameters like robot travel speed, robot jerking, tool orientation and calibration error etc. are not specified effectively. The robot parameters can be considered for smooth welding operation with better weld quality production.
- The optimization of different parameters have not been done yet for obtaining best output in terms of weld quality, cost and welding time with use of sensors in seam tracking. The optimal combination weld parameters, sensor parameters and robot parameters can be obtained by application of different artificial intelligence techniques in process control and monitoring of weld quality.
- The weld quality in a wide range like weld strength, weld depth penetration, weld bead shape and size, uniformity of weld joint about a long length, presence of porosity and cracks etc. need to be explored more with sensor assisted robotic welding.
- The weld-gap or width of weld joint has been taken constant in most of the papers while it may vary depending up on the machining or manufacturing processes of work-piece materials. Again weld-gap is one of the most important weld parameter affecting the weld depth penetration, electrode wire feed rate, maintenance of uniformity of weld joint and robot jerking. Therefore this parameter measurement can be considered for seam tracking operation.
- With the help of sensors, the weld pool geometries can be measured and can be used for simultaneous seam tracking and weld seam inspection.
- From literature survey, it has been found that, for controller design, fuzzy and neural network or simple PID controller have been used. Other advanced soft computing or hybrid optimization techniques can be used for weld parameter control with the seam tracking results obtained from the sensor. The results can be compared further for obtaining better weld quality.

References

- [1] L. Sciacavico, B. Siciliano, *Modelling and Control of Robot Manipulators*, Springer Science & Business Media, 2012.
- [2] E. Abele, M. Weigold, S. Rothenbücher, Modeling and identification of an industrial robot for machining applications, *CIRP Ann.* 56 (1) (2007) 387–390.
- [3] G. Bolmsjö, M. Olsson, P. Cederberg, Robotic arc welding—trends and developments for higher autonomy, *Ind. Robot* 29 (2) (2002) 98–104.
- [4] L. Laiping, C. Shanben, L. Tao, The modeling of welding pool surface reflectance of aluminum alloy pulse GTAW, *Mater. Sci. Eng.* 394 (1) (2005) 320–326.
- [5] J. Pan, A survey of welding sciences in 21st century, *Proceeding of 9th Chinese Welding Conference*, Tianjun, China, vol. 1, (1999), pp. D001–D017.
- [6] S. Chen, T. Qiu, T. Lin, Y. Wu, On intelligentized technologies for modern welding manufacturing, *Chin. J. Mech. Eng.* 16 (4) (2003) 367–370.
- [7] J.N. Pires, A. Loureiro, T. Godinho, P. Ferreira, B. Fernando, J. Morgado, Welding robots, *IEEE Robot. Autom. Mag.* 10 (2) (2003) 45–55.
- [8] S. Chen, Research evolution on intelligentized technologies for robotic welding at SJTU, *Robotic Welding, Intelligence and Automation*, Springer, 2011, pp. 3–14.
- [9] S. Chen, J. Wu, *Intelligentized Technology for Arc Welding Dynamic Process*, Lecture Notes in Electrical and Engineering, LNEE 29 (2008).
- [10] T. Brogårdh, Present and future robot control development an industrial perspective, *Annu. Rev. Control* 31 (1) (2007) 69–79.
- [11] S. Hongyuan, H. Xixia, L. Tao, C. Shanben, Weld formation control for arc welding robot, *Int. J. Adv. Manuf. Technol.* 44 (5) (2009) 512–519.
- [12] T. Chettibi, H. Lehtihet, M. Haddad, S. Hanchi, Minimum cost trajectory planning for industrial robots, *Eur. J. Mech.* 23 (4) (2004) 703–715.
- [13] Y. Xue, I. Kim, J. Son, C. Park, H. Kim, B. Sung, I. Kim, H. Kim, B. Kang, Fuzzy regression method for prediction and control the bead width in the robotic arc-welding process, *J. Mater. Process. Technol.* 164 (2005) 1134–1139.
- [14] S. Chen, T. Qiu, T. Lin, L. Wu, J. Tian, W. Lv, Y. Zhang, Intelligent technologies for robotic welding, *Robot. Weld. Intell. Autom.* (2004) 123–143.
- [15] S. Chen, N. Lv, Research evolution on intelligentized technologies for arc welding process, *J. Manuf. Process* 16 (1) (2014) 109–122.
- [16] J. Xiao-gang, Analysis on arc-welding robot visual control tracking system, *Mach. Des. Manuf.* 5 (2009) 63–65.
- [17] J.N. Pires, A. Loureiro, G. Bölsjö, *Welding Robots: Technology, System Issues and Application*, Springer Science & Business Media, 2006.
- [18] L. Sweet, Sensor-based control systems for arc welding robots, *Robot. Comput. Integr. Manuf.* 2 (2) (1985) 125–133.
- [19] A. Bauchspiess, S.C.A. Alfaro, L.A. Dobrzanski, Predictive sensor guided robotic manipulators in automated welding cells, *J. Mater. Process. Technol.* 109 (1) (2001) 13–19.

- [20] J. Nóbretro Pires, G. Bolmsjö, M. Olsson, Sensors in robotic arc welding to support small series production, *Ind. Robot* 32 (4) (2005) 341–345.
- [21] M. Ushio, W. Mao, Modelling of an arc sensor for dc mig/mag welding in open arc mode: study of improvement of sensitivity and reliability of arc sensors in GMA welding (1st report), *Weld. Int.* 10 (8) (1996) 622–631.
- [22] P. Jiluan, Arc sensing system for automatic weld seam tracking-mathematic model, *Sci. China* 3 (2001) 003.
- [23] H.-S. Moon, S.-H. Ko, J.-C. Kim, Automatic seam tracking in pipeline welding with narrow groove, *Int. J. Adv. Manuf. Technol.* 41 (3) (2009) 234–241.
- [24] M. Fridenfalk, G. Bolmsjö, Design and validation of a universal 6d seam-tracking system in robotic welding using arc sensing, *Adv. Rob.* 18 (1) (2004) 1–21.
- [25] S.-K. Jeong, G.-Y. Lee, W.-K. Lee, S.-B. Kim, Development of high speed rotating arc sensor and seam tracking controller for welding robots, *Industrial Electronics*, 2001. Proceedings. ISIE 2001. IEEE International Symposium on, vol. 2, IEEE, 2001, pp. 845–850.
- [26] H.-j. Chen, H. Zhang, Y. Xiong, Research on welding seam tracking based on HSIC, *Future Intelligent Information Systems*, Springer, 2011, pp. 383–390.
- [27] G. Yanfeng, Z. Hua, Y. Yanhui, Back-stepping and neural network control of a mobile robot for curved weld seam tracking, *Procedia Eng.* 15 (2011) 38–44.
- [28] Arc Sensor by Daihen, USA, (<https://www.daihen-usa.com/product/fd-tr-tig-arc-sensor/>). Accessed: 2018-06-26.
- [29] J. Ge, Z. Zhu, D. He, L. Chen, A vision-based algorithm for seam detection in a paw process for large-diameter stainless steel pipes, *Int. J. Adv. Manuf. Technol.* 26 (9) (2005) 1006–1011.
- [30] H. Ma, S. Wei, Z. Sheng, T. Lin, S. Chen, Robot welding seam tracking method based on passive vision for thin plate closed-gap butt welding, *Int. J. Adv. Manuf. Technol.* 48 (9–12) (2010) 945–953.
- [31] D. Xu, Z. Fang, H. Chen, Z. Yan, M. Tan, Compact visual control system for aligning and tracking narrow butt seams with CO₂ gas-shielded arc welding, *Int. J. Adv. Manuf. Technol.* 62 (9) (2012) 1157–1167.
- [32] L. Nele, E. Sarno, A. Keshari, An image acquisition system for real-time seam tracking, *Int. J. Adv. Manuf. Technol.* 69 (9–12) (2013) 2099–2110.
- [33] Y. Xu, G. Fang, S. Chen, J.J. Zou, Z. Ye, Real-time image processing for vision-based weld seam tracking in robotic GMAW, *Int. J. Adv. Manuf. Technol.* 73 (9–12) (2014) 1413–1425.
- [34] H. Chen, T. Lin, S. Chen, Seam tracking and dynamic process control for high precision arc welding, *Robotic Welding, Intelligence and Automation*, Springer, 2011, pp. 193–201.
- [35] J. Gao, C. Wu, X. Liu, D. Xia, Vision-based weld seam tracking in gas metal arc welding, *Front. Mater. Sci. China* 1 (3) (2007) 268–273.
- [36] L. Zhou, T. Lin, S.-B. Chen, Autonomous acquisition of seam coordinates for arc welding robot based on visual servoing, *J. Intell. Robot. Syst.* 47 (3) (2006) 239–255.
- [37] H. Shen, T. Lin, S. Chen, L. Li, Real-time seam tracking technology of welding robot with visual sensing, *J. Intell. Robot. Syst.* 59 (3–4) (2010) 283–298.
- [38] Y. Xu, G. Fang, N. Lv, S. Chen, J.J. Zou, Computer vision technology for seam tracking in robotic GTAW and GMAW, *Robot Comput. Integr. Manuf.* 32 (2015) 25–36.
- [39] M. Dinham, G. Fang, Autonomous weld seam identification and localisation using eye-in-hand stereo vision for robotic arc welding, *Robot Comput. Integr. Manuf.* 29 (5) (2013) 288–301.
- [40] M. Dinham, G. Fang, The development of a low cost autonomous robotic arc welding system, *Robotic Welding, Intelligence and Automation*, Springer, 2015, pp. 541–550.
- [41] M. Dinham, G. Fang, A low cost hand-eye calibration method for arc welding robots, *Robotics and Biomimetics (ROBIO)*, 2009 IEEE International Conference on, IEEE, 2009, pp. 1889–1893.
- [42] M. Heber, M. Lenz, M. Rüther, H. Bischof, H. Fronthaler, G. Croonen, Weld seam tracking and panorama image generation for on-line quality assurance, *Int. J. Adv. Manuf. Technol.* 65 (9–12) (2013) 1371–1382.
- [43] K. Micallef, G. Fang, M. Dinham, Automatic seam detection and path planning in robotic welding, *Robotic Welding, Intelligence and Automation*, Springer, 2011, pp. 23–32.
- [44] H. Shen, T. Lin, S. Chen, A study on vision-based real-time seam tracking in robotic arc welding, *Robotic Welding, Intelligence and Automation*, Springer, 2007, pp. 311–318.
- [45] L. Xue, L. Xu, Y. Zou, A method of seam tracking based on passive vision, *Robotic Welding, Intelligence and Automation*, Springer, 2011, pp. 131–138.
- [46] H.-C. Kuo, L.-J. Wu, An image tracking system for welded seams using fuzzy logic, *J. Mater. Process. Technol.* 120 (1) (2002) 169–185.
- [47] X. Chen, S. Chen, T. Lin, Recognition of macroscopic seam for complex robotic welding environment, *Robotic Welding, Intelligence and Automation*, Springer, 2007, pp. 171–178.
- [48] M. Dinham, G. Fang, Detection of fillet weld joints using an adaptive line growing algorithm for robotic arc welding, *Robot Comput. Integr. Manuf.* 30 (3) (2014) 229–243.
- [49] F. Shi, L. Zhou, T. Lin, S. Chen, Efficient weld seam detection for robotic welding from a single image, *Robotic Welding, Intelligence and Automation*, Springer, 2007, pp. 289–294.
- [50] B. Li, S. Tan, W. Zhang, A fast GPI line detection method for robot seam tracking, *Robotic Welding, Intelligence and Automation*, Springer, 2011, pp. 57–64.
- [51] X. Gao, S.-J. Na, Detection of weld position and seam tracking based on Kalman filtering of weld pool images, *J. Manuf. Syst.* 24 (1) (2005) 1–12.
- [52] T. Zhang, K. Li, J. Yang, Seam tracking control for mobile welding robot based on vision sensor, *J. Cent. South Univ. Technol.* 17 (6) (2010) 1320–1326.
- [53] Vision Sensor by Motoman, (<https://www.motoman.com/products/vision/> default#related). Accessed: 2018-06-26.
- [54] M. Wilson, The role of seam tracking in robotic welding and bonding, *Ind. Robot* 29 (2) (2002) 132–137.
- [55] M.F. Zaeh, W. Vogl, Interactive laser-projection for programming industrial robots, *Mixed and Augmented Reality*, 2006. ISMAR 2006. IEEE/ACM International Symposium on, IEEE, 2006, pp. 125–128.
- [56] Y. He, Y. Chen, Y. Xu, Y. Huang, S. Chen, Autonomous detection of weld seam profiles via a model of saliency-based visual attention for robotic arc welding, *J. Intell. Robot. Syst.* 81 (3–4) (2016) 395.
- [57] W. Gu, Z. Xiong, W. Wan, Autonomous seam acquisition and tracking system for multi-pass welding based on vision sensor, *Int. J. Adv. Manuf. Technol.* 69 (1–4) (2013) 451–460.
- [58] G. Agapiou, C. Kasiouras, A. Serafetinides, A detailed analysis of the mig spectrum for the development of laser-based seam tracking sensors, *Opt. Laser Technol.* 31 (2) (1999) 157–161.
- [59] P. Xu, G. Xu, X. Tang, S. Yao, A visual seam tracking system for robotic arc welding, *Int. J. Adv. Manuf. Technol.* 37 (1) (2008) 70–75.
- [60] Y. Huang, Y. Xiao, P. Wang, M. Li, A seam-tracking laser welding platform with 3d and 2d visual information fusion vision sensor system, *Int. J. Adv. Manuf. Technol.* 67 (1–4) (2013) 415–426.
- [61] Y. Li, D. Xu, Z. Yan, M. Tan, Girth seam tracking system based on vision for pipe welding robot, *Robotic Welding, Intelligence and Automation*, Springer, 2007, pp. 391–399.
- [62] Z. Xiao, Research on a triline laser vision sensor for seam tracking in welding, *Robotic Welding, Intelligence and Automation*, Springer, 2011, pp. 139–144.
- [63] S. Matsui, G. Goktug, Slit laser sensor guided real-time seam tracking arc welding robot system for non-uniform joint gaps, *Industrial Technology*, 2002. IEEE ICIT'02. 2002 IEEE International Conference on, vol. 1, IEEE, 2002, pp. 159–162.
- [64] M. Fridenfalk, G. Bolmsjö, Design and validation of a universal 6d seam tracking system in robotic welding based on laser scanning, *Ind. Robot* 30 (5) (2003) 437–448.
- [65] K. Sung, H. Lee, Y. Choi, S. Rhee, Development of a multiline laser vision sensor for joint tracking in welding, *Weld. J.* 4 (2009) 79–85.
- [66] D. Chang, D. Son, J. Lee, D. Lee, T.-w. Kim, K.-Y. Lee, J. Kim, A new seam-tracking algorithm through characteristic-point detection for a portable welding robot, *Robot Comput. Integr. Manuf.* 28 (1) (2012) 1–13.
- [67] Q.-Q. Wu, J.-P. Lee, M.-H. Park, C.-K. Park, I.-S. Kim, A study on development of optimal noise filter algorithm for laser vision system in GMA welding, *Procedia Eng.* 97 (2014) 819–827.
- [68] C.-H. Kim, T.-Y. Choi, J. Suh, K.-T. Park, J.-J. Lee, H.-S. Kang, M.-Y. Lee, S.-r. Kim, Sensor guided laser welding robot system, *IFAC Proc. Vol.* 41 (2) (2008) 4324–4329.
- [69] Z. Fang, D. Xu, M. Tan, Visual seam tracking system for butt weld of thin plate, *Int. J. Adv. Manuf. Technol.* 49 (5) (2010) 519–526.
- [70] L. Li, L. Fu, X. Zhou, X. Li, Image processing of seam tracking system using laser vision, *Robotic Welding, Intelligence and Automation*, Springer, 2007, pp. 319–324.
- [71] Z. Yan, D. Xu, Y. Li, M. Tan, A vision-based seam tracking system for submerged arc welding, *Robotic Welding, Intelligence and Automation*, Springer, 2007, pp. 349–357.
- [72] Z. Xiong, W. Wan, J. Pan, Application of fuzzy edge detection in weld seam tracking system, *Robotic Welding, Intelligence and Automation*, Springer, 2011, pp. 323–330.
- [73] Y. He, Y. Xu, Y. Chen, H. Chen, S. Chen, Weld seam profile detection and feature point extraction for multi-pass route planning based on visual attention model, *Robot Comput. Integr. Manuf.* 37 (2016) 251–261.
- [74] J.-P. Lee, Q.-Q. Wu, M.-H. Park, C.-K. Park, I.-S. Kim, A study on optimal algorithms to find joint tracking in GMA welding, *Int. J. Eng. Sci. Innov. Technol.* 3 (2014) 370–380.
- [75] H. Luo, X. Chen, Laser visual sensing and process control in robotic arc welding of titanium alloys, *Robotic Welding, Intelligence and Automation*, Springer, 2004, pp. 110–122.
- [76] J.P. Huissoon, Robotic laser welding: seam sensor and laser focal frame registration, *Robotica* 20 (03) (2002) 261–268.
- [77] W. Huang, R. Kovacevic, Development of a real-time laser-based machine vision system to monitor and control welding processes, *Int. J. Adv. Manuf. Technol.* 63 (1–4) (2012) 235–248.
- [78] H. Luo, X. Chen, Laser visual sensing for seam tracking in robotic arc welding of titanium alloys, *Int. J. Adv. Manuf. Technol.* 26 (9) (2005) 1012–1017.
- [79] W. Zhang, Z. Zheng, X. Ma, Q. Chen, D. Du, Circular sub-window multi-step GPI method in seam tracking of welding robot based on 3d vision, *Intell. Robot. Appl.* (2008) 916–926.
- [80] D. Xu, M. Tan, X. Zhao, Z. Tu, Seam tracking and visual control for robotic arc welding based on structured light stereovision, *Int. J. Autom. Comput.* 1 (1) (2004) 63–75.
- [81] Y. Quan, Q. Bi, Tracking and monitoring of 3-dimensions of welding seam and width in fillet welding of corrugated sheet, *Int. J. Control Autom.* 8 (5) (2015) 337–350.
- [82] K. Haug, G. Pritschow, Robust laser-stripe sensor for automated weld-seam-tracking in the shipbuilding industry, *Industrial Electronics Society*, 1998. IECON'98. Proceedings of the 24th Annual Conference of the IEEE, vol. 2, IEEE, 1998, pp. 1236–1241.
- [83] P. Kim, S. Rhee, C.H. Lee, Automatic teaching of welding robot for free-formed seam using laser vision sensor, *Opt. Lasers Eng.* 31 (3) (1999) 173–182.
- [84] J.S. Kim, Y.T. Son, H.S. Cho, K.I. Koh, A robust visual seam tracking system for robotic arc welding, *Mechatronics* 6 (2) (1996) 141–163.

- [85] W.-S. Yoo, S.-J. Na, Determination of 3-d weld seams in ship blocks using a laser vision sensor and a neural network, *J. Manuf. Syst.* 22 (4) (2003) 340–347.
- [86] G. Pritschow, G. Mueller, H. Horber, Fast and robust image processing for laser stripe-sensors in arc welding automation, *Proc. IEEE Int. Symp. Ind. Electron.*, vol. 2, (2002), pp. 651–656.
- [87] H. Koch, A. Konig, K. Kleinmann, A. Weigl-Seitz, J. Suchy, Predictive robotic contour following using laser-camera-triangulation, *Advanced Intelligent Mechatronics (AIM)*, 2011 IEEE/ASME International Conference on, IEEE, 2011, pp. 422–427.
- [88] Laser Vision Sensor by Scansonic, (<https://www.scansonic.de/en/products/th6d-optical-sensor>). Accessed: 2018-06-27.
- [89] E.L. Estochen, C.P. Neuman, F.B. Prinz, Application of acoustic sensors to robotic seam tracking, *IEEE Trans. Ind. Electron.* (3) (1984) 219–224.
- [90] T.F. Bastos, J.M. Martin, L. Calderon, R. Ceres, Weld seams detection and recognition for robotic arc-welding through ultrasonic sensors, *Industrial Electronics*, 1994. Symposium Proceedings, ISIE'94., 1994 IEEE International Symposium on, IEEE, 1994, pp. 310–315.
- [91] B. Maqueira, C.I. Umeagukwu, J. Jarzynski, Application of ultrasonic sensors to robotic seam tracking, *IEEE Trans. Robot. Autom.* 5 (3) (1989) 337–344.
- [92] S. Zhang, Y. Zhang, R. Kovacevic, Noncontact ultrasonic sensing for seam tracking in arc welding processes, *J. Manuf. Sci. Eng.* 120 (1998) 600–608.
- [93] B. You, J. Kim, A study on an automatic seam tracking system by using an electromagnetic sensor for sheet metal arc welding of butt joints, *Proc. Inst. Mech. Eng. Part B* 216 (6) (2002) 911–920.
- [94] X. Gao, D. You, S. Katayama, Infrared image recognition for seam tracking monitoring during fiber laser welding, *Mechatronics* 22 (4) (2012) 370–380.
- [95] S. Nagarajan, P. Banerjee, W. Chen, B.A. Chin, Control of the welding process using infrared sensors, *IEEE Trans. Robot. Autom.* 8 (1) (1992) 86–93.
- [96] P. Banerjee, S. Govardhan, H. Wikle, J. Liu, B. Chin, Infrared sensing for on-line weld geometry monitoring and control, *J. Eng. Ind.* 117 (3) (1995) 323–330.
- [97] H. Wikle, S. Kottilingam, R. Zee, B. Chin, Infrared sensing techniques for penetration depth control of the submerged arc welding process, *J. Mater. Process. Technol.* 113 (1) (2001) 228–233.
- [98] Y. Xu, N. Lv, J. Zhong, H. Chen, S. Chen, Research on the real-time tracking information of three-dimension welding seam in robotic GTAW process based on composite sensor technology, *J. Intell. Robot. Syst.* 68 (2) (2012) 89–103.
- [99] S. Presern, L. Gyergyek, An intelligent tactile sensor-an on-line hierarchical object and seam analyzer, *IEEE Trans. Pattern Anal. Mach. Intell.* (2) (1983) 217–220.
- [100] K. Suwanratchatamane, M. Matsumoto, S. Hashimoto, A novel tactile sensor torch system for robot manipulator and active object edge tracking, *Industrial Electronics*, 2008. IECON 2008. 34th Annual Conference of IEEE, IEEE, 2008, pp. 2617–2622.
- [101] K. Suwanratchatamane, M. Matsumoto, S. Hashimoto, Robotic tactile sensor system and applications, *IEEE Trans. Ind. Electron.* 57 (3) (2010) 1074–1087.
- [102] Tactile Sensor by Lincolnelectric, (<https://www.lincolnelectric.com/en-us/support/process-and-theory/pages/tactile-seam-tracking-systems.aspx>). Accessed: 2018-06-27.
- [103] Touch Sensor by Daihen, USA, (<https://www.daihen-usa.com/product/fd-wd-wire-detection/>). Accessed: 2018-06-27.
- [104] K. Pietrzak, S. Packer, Vision-based weld pool width control, *J. Eng. Ind.* 116 (1994). 86–86
- [105] Y. Zhang, R. Kovacevic, S. Ruan, Sensing and control of weld pool geometry for automated GTA welding, *Trans. ASME* 117 (1995) 210–222.
- [106] R. Kovacevic, Y. Zhang, Real-time image processing for monitoring of free weld pool surface, *J. Manuf. Sci. Eng.* 119 (2) (1997) 161–169.
- [107] O. Madsen, H. Shen, H. Ma, T. Lin, S. Chen, Research on weld pool control of welding robot with computer vision, *Ind. Robot* 34 (6) (2007) 467–475.
- [108] S. Pal, S.K. Pal, A.K. Samantaray, Artificial neural network modeling of weld joint strength prediction of a pulsed metal inert gas welding process using arc signals, *J. Mater. Process. Technol.* 202 (1) (2008) 464–474.
- [109] S.-B. Chen, J. Wu, Real-time control of weld pool dynamics during robotic GTAW, *Intelligentized Methodology for Arc Welding Dynamical Processes*, Springer, 2009, pp. 221–273.
- [110] H. Ma, S. Wei, T. Lin, S. Chen, L. Li, Binocular vision system for both weld pool and root gap in robot welding process, *Sensor Rev.* 30 (2) (2010) 116–123.
- [111] T.L. Chung, T.H. Bui, T.T. Nguyen, S.B. Kim, Sliding mode control of two-wheeled welding mobile robot for tracking smooth curved welding path, *J. Mech. Sci. Technol.* 18 (7) (2004) 1094–1106.
- [112] J. Liu, Y. Hu, B. Wu, K. Zhou, M. Ge, An adaptive sliding mode control technology for weld seam tracking, *Front. Mech. Eng.* 10 (1) (2015) 95–101.
- [113] B. Deepak, C. Rao, B. Raju, Weld seam tracking and simulation of 3-axis robotic arm for performing welding operation in cad environment, *CAD/CAM, Robotics and Factories of the Future*, Springer, 2016, pp. 405–415.
- [114] P. Neto, N. Mendes, R. Araújo, J. Norberto Pires, A. Paulo Moreira, High-level robot programming based on cad: dealing with unpredictable environments, *Ind. Robot* 39 (3) (2012) 294–303.
- [115] G. Carvalho, M. Siqueira, S. Absi-Alfaro, Off-line programming of flexible welding manufacturing cells, *J. Mater. Process. Technol.* 78 (1) (1998) 24–28.